

# What do Boards Know? The Value of CEO Selection and Monitoring\*

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## Abstract

How much do boards know about their CEOs, and what is that information worth? We estimate a dynamic model of CEO succession in which boards learn about incumbent quality from noisy performance and screen internal candidates before promoting them. Boards' information is valuable but limited: screening resolves a third of the uncertainty about internal candidates, and a year of performance 3.8 percent of that about the incumbent. What boards know is worth 4.6 percent of firm value at selection and 4.1 percent at monitoring, and full information would raise firm value by 19 percent. Insiders' information advantage explains both why two-thirds of new CEOs come from within and why these insiders outperform at appointment.

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# 1 Introduction

When a chief executive’s job comes open, the board must decide whom to appoint. While a chief executive is in place, it must decide whether to keep the incumbent. Both decisions turn on what the board knows about the quality of the executive in question, and neither can be made better than that knowledge allows. Yet the two have been studied very unevenly. Beginning with Weisbach (1988), the sensitivity of forced turnover to firm performance has served as a standard measure of whether a board is doing its job, and a large literature has refined it: how it has changed over time (Huson et al., 2001; Kaplan and Minton, 2012), whether boards credit industry luck (Jenter and Kanaan, 2015), and how much of it is obscured by the classification of departures (Jenter and Lewellen, 2020). There is no comparable statistic for the appointment decision: we can say how well boards dismiss, but not how well they choose. This asymmetry would matter less if the neglected margin were the smaller one. It is not obviously so. Roughly two-thirds of chief executives in our sample are promoted from within, and the insiders boards select outperform outside hires.

Neither margin reveals what boards know. On the appointment side, what a board learns about a candidate before choosing is never observed. Cziraki and Jenter (2021) document that the market for chief executives is dominated by internal promotion without evidence that insiders are intrinsically superior, and conjecture that boards simply know more about the candidates they have employed. He and Schroth (2024) ask directly why external successions are so rare, and Cvijanović et al. (2023) study the succession planning through which boards prepare for them. None of this work measures how much boards know about the candidates they appoint. On the retention side, the statistic that does exist is the product of two things: how much a year of performance moves the board’s assessment of its chief executive, and how readily the board acts on a given assessment. The first is information; the second is not. Engel et al. (2003) show that the turnover-performance relation varies with the informativeness of the performance measure, and Bushman et al. (2010) document a related dependence on noise. A board that rarely dismisses a poor performer may be a board that tolerates one, or a board that cannot yet tell.

This paper separates the two, and measures board information at both margins: at *selection*, what a board can tell about a candidate before appointing one, and at *monitoring*, what it learns about the executive it already has. We specify and estimate a dynamic model of chief executive retention and succession in which the precision of the board’s information enters as a small number of parameters. A board employs one chief executive whose productivity is persistent but evolving and which it never observes directly. Each period it learns about the incumbent from profitability, which confounds the executive’s contribution with

industry conditions, and from a noisy benchmark of those conditions, so that the degree to which industry luck is credited to the chief executive is itself estimated rather than assumed. When the job comes open, the board chooses between an insider, who inherits part of the predecessor’s trajectory and about whom the board observes a signal before appointing, and an outsider, drawn from the market, about whom it observes no comparable signal. Boards need not act on firm value alone: we allow a non-pecuniary cost of dismissal and a non-pecuniary preference for internal appointment, so that favouritism is given a chance to explain the prevalence of internal promotion. We estimate the model by Simulated Method of Moments on a panel of 3,158 firms and 6,508 chief executive spells from 1995 to 2019.

Boards’ information turns out to be real but limited, and limited in different ways at the two margins. The pre-appointment signal resolves roughly a third of the uncertainty about an internal candidate’s productivity, and the confidence interval rejects both extremes: the signal is neither uninformative nor fully revealing. On-the-job learning is slow: a full year of information resolves only 3.8% of the board’s uncertainty about its chief executive’s current productivity, a pace consistent with the gradual resolution of uncertainty documented by Taylor (2013) and Pan et al. (2015). Boards filter roughly 65% of industry performance out of their assessments. Among the parameters that do not concern information, the non-pecuniary cost of dismissal is large and precisely estimated, while the preference for internal appointment cannot be distinguished from zero.

The central results price the board’s information at each margin. Replacing the board’s choice of successor with a coin flip at the empirical insider share, holding the composition of appointments and the information structure fixed, reduces firm value by 4.6%. Replacing the board’s dismissal rule with uninformed dismissal at the empirical rate reduces firm value by 4.1%, and destroying both costs 9.2%. Boards’ use of what they know is therefore worth about as much when they choose a chief executive as when they judge one, though only the dismissal decision has a standard measure behind it.

Asking instead how far boards stand from full information changes the question. Endowing boards with perfect knowledge of chief executive productivity would raise firm value by 18.8%. Strikingly, either margin delivers most of that on its own: perfect selection alone is worth 17.1% and perfect monitoring alone 12.6%. The two are therefore close substitutes, because a board that always appoints the better candidate has little left to learn about the executive it installed, and a board that can identify a failing executive at once has less need to have screened beforehand. So while boards’ existing information is worth about the same at the two margins, the value they are still leaving on the table is larger at selection.

Perfect filtering of industry conditions, by contrast, contributes 0.05 percentage points. Boards do respond to industry conditions in their dismissal decisions, in our data as elsewhere

(Kaplan and Minton, 2012; Jenter and Kanaan, 2015), and the informativeness principle of Holmström (1979) implies they should not. The tension has generated an active literature (Gibbons and Murphy, 1990; Bertrand and Mullainathan, 2001). But the value at stake in that behaviour is smaller than the value at stake in the selection of candidates by more than two orders of magnitude.

The estimates also speak to a long debate about whether insiders or outsiders make better chief executives (Shen and Cannella, 2002; Zhang and Rajagopalan, 2010; Quigley et al., 2019). In our estimates the two candidate pools are nearly identical: the long-run productivity of internal appointees exceeds that of outsiders by 0.6% of assets, against a cross-sectional dispersion of executive productivity of 3.5%. The observed advantage of insiders arises at appointment, from what boards know about internal candidates and from the successful regimes those candidates inherit. Under full information the productivity gap between appointed insiders and appointed outsiders narrows by about three-quarters, and the share of appointments filled internally falls from two-thirds to 0.58. The insider advantage is therefore primarily an information advantage rather than an endowment of the internal pipeline. The interpretation has theoretical precedent: Ferreira and Nikolowa (2023) show that when employers know their own managers better than the market does, promoted insiders outperform external hires even though both are drawn from a common talent distribution. What we add is the magnitude, and the finding that for chief executives the informational channel accounts for the observed gap essentially on its own. Consistent with this reading, removing the board’s pre-appointment signal raises the insider share to 0.86 while cutting the average productivity of appointed insiders by more than half: unable to distinguish among internal candidates, boards promote indiscriminately. Better information makes boards more selective, not more insider-prone.

Our paper contributes to three literatures. The first is the structural analysis of chief executive turnover. Taylor (2010) measures what boards’ reluctance to dismiss a chief executive costs shareholders; in his framework a departing executive is replaced by a random draw from the population, so there is no appointment decision to price. We measure what boards’ ability to choose is worth and find the two comparable in magnitude. Taylor (2013) and Pan et al. (2015) infer the pace of board learning from compensation dynamics and from the decline in return volatility over tenure; we infer it from the personnel decisions themselves, and recover in addition the share of uncertainty resolved before the appointment. A related structural literature estimates models of executive careers in which the object of interest is the compensation contract (Margiotta and Miller, 2000; Gayle and Miller, 2009; Gayle et al., 2015). Our board writes no contract; what we recover is the precision of its information rather than the incentives it provides.

The second is the empirical literature on boards. Analyzing the minutes of board meetings, Schwartz-Ziv and Weisbach (2013) find that directors devote most of their attention to supervising management, and that the appointment or dismissal of an executive is the issue on which boards most often take an active hand. By separating what boards know from how they act on it, we show that the turnover-performance sensitivity understates board effectiveness to the extent that dismissal is limited by information rather than by will. We also supply a measure for the other half of the board’s personnel problem, the appointment decision, for which no comparable statistic exists. Our estimates also bear on the interpretation of internal promotion. The prevalence of insiders has been read as evidence of favouritism or entrenchment (Agrawal et al., 2006; Mobbs and Raheja, 2012); we cannot distinguish the non-pecuniary preference for internal appointment from zero, in a model where it was free to be large, and attribute the pattern instead to what boards know.

The third is the literature on internal labour markets and employer learning. That firms know more about the workers they already employ than about outside applicants is a long-standing idea (Greenwald, 1979; Lazear, 1998; DeVaro and Morita, 2013), and the informational advantage boards hold over internal candidates is the accumulated product of having observed those candidates in junior roles (Baker et al., 1994). Ferreira and Nikolowa (2023) study this idea in equilibrium: employers who have discovered talent internally can retain it against a market that cannot verify it, so promotion from within prevails and appointments from outside are comparatively rare. Our estimates put a value on that accumulated knowledge at the top of the organization: the option to select on privately observed fit, which is what distinguishes an internal promotion from an external hire (Chan, 1996; Bidwell, 2011), is among the most valuable features of the succession process. That literature also asks how fast such knowledge accumulates (Altonji and Pierret, 2001). Employers’ initial expectation errors about a young worker fall by half within three years (Lange, 2007), whereas boards resolve only sixteen percent of their initial uncertainty about a new chief executive over the same period, a difference consistent with how noisy firm performance is as a signal of any one executive’s contribution. This has a practical counterpart in succession planning, whose disclosure has itself been the subject of governance attention (McConnell and Qi, 2022; Cvijanović et al., 2023), and which a recent literature takes as its subject: Celentano and Mello (2024) ask why firms so often lack a succession plan at all, and Geelen and Hajda (2024) study how the anticipation of succession shapes firm policy. The precision of the board’s pre-appointment signal is, in effect, what such planning produces.

The remainder of the paper proceeds as follows. Section 2 describes the data and the facts that motivate and discipline the model. Section 3 presents the model. Section 4 discusses identification and estimation. Section 5 reports the estimates. Section 6 prices the board’s

information through a series of counterfactuals. Section 7 concludes.

## 2 Data and Stylized Facts

We begin by describing our sample and documenting the features of CEO succession that motivate and discipline our model. Internal promotion dominates the market for CEOs: 66.2% of the successions in our sample are filled from within the firm (Cziraki and Jenter, 2021). What the board knows when it makes that choice is never observed, but it leaves traces in what boards do and in what follows. The insider advantage is concentrated at appointment and fades over the spell, rather than persisting as a constant gap. And boards weigh recent performance above older performance not only when they dismiss a chief executive, as Jenter and Lewellen (2020) document, but also when they choose between internal and external candidates.

**Sample construction.** We construct a panel of publicly traded firms spanning 1995 to 2019 from three sources of information. ExecuComp provides CEO-level information for S&P 1500 firms, including executives’ prior experience at the firm, tenure, and turnover. Compustat provides firm fundamentals. We use two measures of performance. The first is operating ROA: operating income before depreciation scaled by average total assets. It appears both raw and industry-adjusted; the adjustment subtracts the industry-year mean computed over the universe of Compustat firms.<sup>1</sup> The second is the market-adjusted stock return, which comes from CRSP. Lastly, the CEO dismissal database of Gentry et al. (2021) allows us to classify CEO departures as either forced or voluntary.<sup>2</sup> Following Parrino (1997), we classify CEOs as *insiders* or *outsiders* using their prior tenure with the firm at the time of appointment: insiders are CEOs who are founders or who have more than one year of prior experience at the firm, and all other CEOs are classified as outsiders.

Our final sample consists of 38,850 firm-year observations covering 3,158 distinct firms and 6,508 CEO spells. Roughly two-thirds of firm-years are insider-led. The sample contains 3,463 CEO successions, of which 66.2% result in an internal appointment. The predominance of internal promotion is a stable feature of the market for CEOs: comparable insider shares have been documented by Zhang and Rajagopalan (2010), Cremers and Grinstein (2014), and Cziraki and Jenter (2021), with external hiring concentrated among smaller firms and declining with firm size.

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<sup>1</sup>Industries are GICS industry groups; Appendix A provides details on the construction of the industry benchmark.

<sup>2</sup>The database codes the circumstances of each departure of an S&P 1500 chief executive from contemporaneous press reports; Appendix A provides details.

Table 1 reports summary statistics separately for insider- and outsider-led firm-years. Three patterns are worth noting. First, insider-led firm-years display higher profitability: operating ROA averages 12.7% against 10.8% for outsider-led firm-years, and a gap of half a percentage point survives industry adjustment. Second, insider-led firms are substantially larger, with roughly twice the assets and revenue of outsider-led firms. Third, the dispersion of profitability is markedly higher under outsider CEOs, a standard deviation of industry-adjusted ROA of 13.5% against 11.3%, while market-adjusted returns display no insider advantage. The higher mean and lower variance of performance under insider CEOs are consistent with long-standing conjectures that outsider appointments are draws from a riskier pool (Parrino, 1997; Hermalin, 2005; Chen and Hambrick, 2012). The advantage does not appear in returns, as one would expect if an anticipated difference in performance is already reflected in the price.

Table 1: Summary Statistics by CEO Type

|                                 | Outsider-led |         | Insider-led |         | Difference |
|---------------------------------|--------------|---------|-------------|---------|------------|
|                                 | Mean         | SD      | Mean        | SD      |            |
| Operating ROA                   | 0.108        | (0.133) | 0.127       | (0.105) | +0.019***  |
| Industry-adjusted operating ROA | 0.064        | (0.135) | 0.069       | (0.113) | +0.005***  |
| Market-adjusted return          | 0.066        | (0.534) | 0.054       | (0.449) | −0.012**   |
| Assets (\$ billions)            | 10.4         | (66.0)  | 21.5        | (112.7) | +11.1***   |
| Revenue (\$ billions)           | 3.5          | (10.0)  | 7.4         | (22.7)  | +3.9***    |
| Observations (firm-years)       | 13,100       |         | 25,047      |         |            |

Notes: Firm-year observations, split by whether the current CEO was appointed from within the firm. Difference is the insider mean minus the outsider mean; significance is assessed with unequal-variance  $t$ -tests. Assets and revenue are in billions of nominal U.S. dollars. Firm-years for which the CEO's type cannot be classified (703) are excluded from the split. \* $p < .1$ , \*\* $p < .05$ , \*\*\* $p < .01$ .

**Stylized facts.** Throughout, we decompose performance into a firm-specific and an industry component: the industry component is the industry-year mean, and the firm-specific component is the deviation from it, which for profitability is the industry-adjusted measure of Table 1. Three regressors summarize a CEO's record: *recent* performance (the one-year lag of the firm-specific component), *older* performance (the mean of the two- and three-year lags), and the *industry* component (its one-year lag). We measure performance both with market-adjusted stock returns and with industry-adjusted ROA, and we standardize all covariates, so that coefficients measure the change in probability per standard deviation of the regressor.

Table 2 reports linear probability models of forced turnover. Two patterns stand out. First, termination is considerably more sensitive to recent than to older performance (Jenter and Lewellen, 2020): a one-standard-deviation improvement in recent returns lowers the probability of termination by roughly one percentage point, a third of the baseline rate, with older performance carrying about half that weight. Second, termination also responds to the industry component of performance (Jenter and Kanaan, 2015), at roughly 0.3 to 0.4 percentage points per standard deviation, well below the firm-specific sensitivity but not zero. Insider CEOs are 1.4 percentage points less likely to be fired than outsiders, though the gap disappears within firm, indicating that it reflects the composition of firms that promote internally rather than differential treatment.

Table 2: Forced Turnover and the CEO's Record

|                             | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
|-----------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| Recent performance, ROA     | −0.017***<br>(0.003) |                      | −0.012***<br>(0.003) | −0.019***<br>(0.004) |                      | −0.014***<br>(0.004) |
| Older performance, ROA      | 0.010***<br>(0.003)  |                      | 0.006**<br>(0.003)   | 0.007**<br>(0.003)   |                      | 0.005*<br>(0.003)    |
| Industry, ROA               | −0.003**<br>(0.001)  |                      | −0.003**<br>(0.001)  | −0.006<br>(0.005)    |                      | −0.004<br>(0.005)    |
| Recent performance, returns |                      | −0.012***<br>(0.001) | −0.011***<br>(0.001) |                      | −0.011***<br>(0.002) | −0.010***<br>(0.002) |
| Older performance, returns  |                      | −0.003**<br>(0.002)  | −0.002<br>(0.002)    |                      | −0.007***<br>(0.002) | −0.005***<br>(0.002) |
| Industry, returns           |                      | −0.002<br>(0.001)    | −0.002<br>(0.001)    |                      | −0.004**<br>(0.001)  | −0.004**<br>(0.001)  |
| Insider                     | −0.014***<br>(0.003) | −0.015***<br>(0.003) | −0.015***<br>(0.003) | 0.011<br>(0.008)     | 0.011<br>(0.008)     | 0.011<br>(0.008)     |
| Firm + year FE              | No                   | No                   | No                   | Yes                  | Yes                  | Yes                  |
| Observations                | 20,811               |                      |                      |                      |                      |                      |

Notes: Linear probability models of forced CEO turnover on standardized components of the CEO's performance record. Recent and older performance are the firm-specific components of one-year-lagged and two-to-three-year-lagged performance; the industry regressor is the lagged industry-year mean. Standard errors clustered by firm in parentheses. \* $p < .1$ , \*\* $p < .05$ , \*\*\* $p < .01$ .

Table 3 turns to the hiring margin, where we apply the same specification to the choice of successor. The sample conditions on CEO turnover, and the dependent variable indicates that the successor was promoted from within. The probability of internal promotion rises with the *departing* CEO's performance, again more strongly with recent than with older performance: a one-standard-deviation improvement in recent returns raises the probability of an internal appointment by roughly six percentage points. It rises with the industry component of that record as well, by about five percentage points per standard deviation

of industry ROA, though not within firm. The hiring and firing margins thus respond symmetrically to the same information: a record that would have kept the incumbent also promotes her deputy. The pattern is consistent with boards treating a successful record as evidence that the incumbent regime works, and promoting from within to preserve it, while a poor record argues for the fresh draw an outsider represents. The symmetry also carries a subtler implication: whatever shapes the board’s assessment of the departing CEO, including any component of performance the board fails to attribute correctly, propagates into the choice of who runs the firm next.

Table 3: Internal Promotion and the Departing CEO’s Record

|                             | (1)                 | (2)                 | (3)                 | (4)                | (5)                 | (6)                 |
|-----------------------------|---------------------|---------------------|---------------------|--------------------|---------------------|---------------------|
| Recent performance, ROA     | 0.061***<br>(0.018) |                     | 0.027<br>(0.019)    | 0.060**<br>(0.030) |                     | 0.024<br>(0.031)    |
| Older performance, ROA      | −0.018<br>(0.019)   |                     | 0.003<br>(0.019)    | −0.012<br>(0.030)  |                     | 0.003<br>(0.031)    |
| Industry, ROA               | 0.052***<br>(0.011) |                     | 0.049***<br>(0.011) | 0.031<br>(0.036)   |                     | 0.036<br>(0.037)    |
| Recent performance, returns |                     | 0.071***<br>(0.008) | 0.067***<br>(0.008) |                    | 0.058***<br>(0.013) | 0.056***<br>(0.013) |
| Older performance, returns  |                     | 0.031***<br>(0.009) | 0.028***<br>(0.009) |                    | 0.029**<br>(0.013)  | 0.025*<br>(0.014)   |
| Industry, returns           |                     | 0.019**<br>(0.008)  | 0.016*<br>(0.008)   |                    | 0.001<br>(0.014)    | −0.002<br>(0.014)   |
| Firm + year FE              | No                  | No                  | No                  | Yes                | Yes                 | Yes                 |
| Observations                |                     |                     |                     | 2,791              |                     |                     |

Notes: Linear probability models of an internal appointment, estimated on succession events. The dependent variable equals one if the incoming CEO was promoted from within the firm; regressors are the standardized components of the *departing* CEO’s performance record, defined as in Table 2. Standard errors clustered by firm in parentheses. \* $p < .1$ , \*\* $p < .05$ , \*\*\* $p < .01$ .

If boards promote insiders after strong records, do the insiders they promote perform better? Figure 1 plots the insider performance premium by tenure, estimated from a regression of industry-adjusted ROA on CEO type interacted with tenure bins, with firm and year effects. Newly appointed insiders outperform newly appointed outsiders by 2.2 percentage points of ROA in their first year, but the premium fades quickly, and is statistically indistinguishable from zero from the fourth year onward. The fade is informative about *why* insiders outperform. If insiders were simply better managers, the gap should persist at every tenure. If instead their advantage derives from the circumstances of their appointment, the information boards hold about them and the successful regimes they inherit, the gap should decay as those circumstances depreciate. The profile favours the second: the insider advantage looks like a feature of how insiders are chosen and what they inherit, rather than a durable differ-

ence between the candidate pools. The reduced form cannot rule out differential selection over tenure, and the structural model separates these forces.

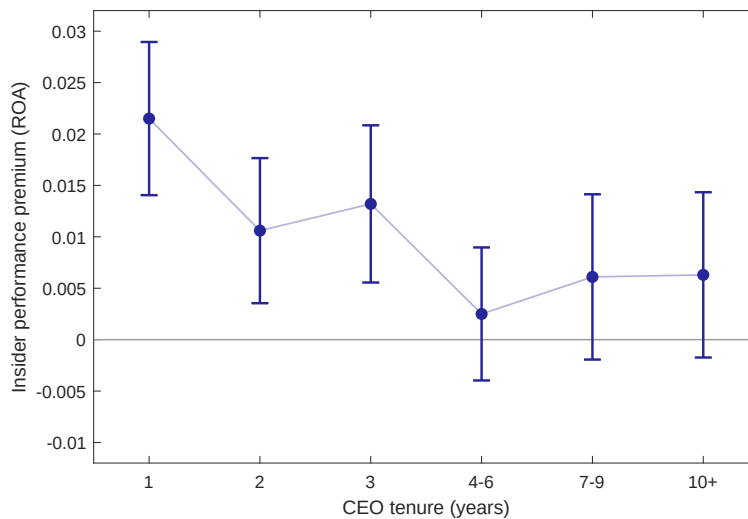


Figure 1: Insider Performance Premium over Tenure

Notes: Point estimates and 95% confidence intervals for the coefficients on insider  $\times$  tenure-bin indicators from a regression of industry-adjusted ROA on CEO type interacted with tenure, with firm and year effects; standard errors are clustered by firm. The regression also includes an indicator for the years surrounding a succession; its coefficient is  $-0.0048$  ( $0.0030$ ).

A final fact bears directly on what boards know about the candidates they appoint. Among CEO spells within the same firm, the between-CEO dispersion of average performance is two-thirds larger for outsider than for insider CEOs ( $0.0038$  versus  $0.0023$ ). The insiders boards promote are, in this respect, a more homogeneous group than the outsiders they hire.

Taken together, the facts admit more than one reading. That boards promote internally after strong records, and that the insiders they promote outperform outsiders at appointment, may reflect what boards know about internal candidates before appointing them, or genuine differences between the internal and external talent pools. The prevalence of internal promotion may also reflect performance continuity, or a taste for in-house management unrelated to firm value. Separating these explanations requires a model that can reproduce the patterns documented here: the weight boards place on recent performance relative to older, the symmetry of the hiring and firing margins, an insider advantage concentrated at appointment and fading thereafter, and a promoted pool markedly less dispersed than a hired one. Section 3 specifies such a model.

### 3 A Dynamic Model of CEO Turnover

A board employs one CEO whose productivity, evolving but partially persistent, it never observes directly. Instead, it learns about the productivity of its CEO from firm profitability, which confounds the executive's contribution with industry conditions, and from a separate, noisy signal of those conditions (such as analyst commentary on industry trends). At succession the board chooses between an insider, who can be screened beforehand and who partially inherits the predecessor's productivity trajectory, and an unscreened outsider whose productivity trajectory is independent of prior firm management. Additional incentives shape turnover decisions beyond information: the board faces pecuniary and non-pecuniary costs of firing and a non-pecuniary preference for internal appointment, which captures insider favoritism.

At each time  $t$ , firm  $j$  in industry  $n$  employs a single CEO  $i$ . We let  $\tau_{it} \in \mathbb{Z}_+$  denote the executive's tenure,  $\theta_{it} \in \mathbb{R}$  denote their productivity, and  $h_i \in \{I, O\}$  denote whether their appointment was internal or external. CEOs can depart voluntarily,  $v_{jt} \in \{0, 1\}$ , or they can be fired,  $d_{jt}^f \in \{0, 1\}$ . At the beginning of each period, voluntary separation is realized. If the CEO retires ( $v_{jt} = 1$ ) the board must hire a replacement. If the CEO does not retire ( $v_{jt} = 0$ ) the board has the option to retain ( $d_{jt}^f = 0$ ) or fire ( $d_{jt}^f = 1$ ) their CEO. Upon turnover, the board chooses to appoint a new executive from the pool of internal or external talent, and it always has one candidate available from each pool.

**Firm profitability.** The profitability of firm  $j$  is given by:

$$Y_{jt} = \iota_j + y_{jt} - c(v_{jt} + d_{jt}^f) \quad (1)$$

where  $\iota_j \sim N(\bar{\iota}, \sigma_\iota^2)$  is a permanent firm component,  $y_{jt}$  is the time-varying component of profitability, and  $c$  is the monetary cost of CEO turnover, capturing the financial cost of executive search, severance payouts, and organizational disruptions to the firm that occur during managerial transitions. The time-varying component of profitability is persistent and depends on CEO ability and an industry factor according to the following mean-reverting process:

$$y_{jt} = \rho y_{jt-1} + (1 - \rho) \theta_{it} + g_{nt} + \varepsilon_{jt} \quad (2)$$

where  $\rho$  captures persistence in firm performance,  $g_{nt}$  is the profitability factor common to all firms in industry  $n$ , and  $\varepsilon_{jt} \sim N(0, \sigma_\varepsilon^2)$  is an idiosyncratic profitability shock. The industry factor is itself persistent such that:

$$g_{nt} = \phi_g g_{nt-1} + \zeta_{nt} \quad (3)$$

where  $\phi_g$  captures persistence in industry-level profitability and where  $\zeta_{nt} \sim N(0, \sigma_\zeta^2)$  are industry innovations. Through its dependence on prior profitability, current profitability  $y_{jt}$  is affected by CEO productivity and shocks from prior periods. Thus, holding the current executive’s productivity constant, firms where the new CEO succeeds a high-productivity one will enjoy higher performance on average.

**CEO productivity and succession.** The executive’s impact on firm profitability is not fixed and it evolves according to:

$$\theta_{it} = \mu_{h_i} + \phi(\theta_{it-1} - \mu_{h_i}) + \eta_{it} \quad (4)$$

where  $\mu_h$  is a type-specific long-run mean capturing population differences between internal and external talent,  $\phi$  governs the persistence of executive productivity over their tenure at the firm, and  $\eta_{it} \sim N(0, \sigma_\eta^2)$  is an innovation to CEO productivity. We normalize the outsider mean to zero,  $\mu_O \equiv 0$ , and leave  $\mu_I$  as a free parameter capturing the effect of firm-specific human capital accumulated at lower ranks within the firm on the productivity of insider appointees (Murphy and Zabojnik, 2006; Custódio et al., 2013; Frydman, 2019). While we refer to  $\theta_{it}$  as CEO productivity, under our specification  $\theta_{it}$  can be understood more broadly as the quality of the firm’s leadership under CEO  $i$ : it evolves as implemented strategies succeed, adapt, and decay, rather than remaining a fixed trait of any executive. The specification in (4) spans the two modeling approaches of managerial quality in the turnover literature: as  $\phi \rightarrow 1$ , productivity becomes a fixed attribute that the board learns over time, as in Jovanovic (1979) and Taylor (2010), while  $\phi < 1$  allows leadership quality to drift over an executive’s tenure, consistent with evidence that both persistent managerial styles (Bertrand and Schoar, 2003) and evolving executive–firm match quality (Allgood and Farrell, 2003; Pan, 2017) shape performance.

The executive’s productivity at the beginning of their tenure, in other words the initial condition of their productivity trajectory at the firm, depends on whether they are hired externally or internally. The outsider candidate is a fresh draw from the stationary type- $O$  productivity distribution centered at zero with stationary variance  $\sigma_\theta^2 = \sigma_\eta^2/(1 - \phi^2)$  implied by (4). The insider candidate, by contrast, inherits part of the organizational trajectory of the predecessor. We capture this organizational continuity through the joint distribution of productivity for the two executives involved in an internal succession: the departing CEO and the internal candidate are drawn from the stationary distribution of their own type, and the two draws are jointly normal with correlation  $\lambda$ . Hence, conditioning on the predecessor’s

productivity  $\theta_{it}$ , the internal candidate's productivity  $\theta_{i't}$  is given by:<sup>3</sup>

$$\theta_{i't} = \mu_I + \lambda(\theta_{it} - \mu_{h_i}) + \chi_{i'}, \quad \chi_{i'} \sim N(0, \sigma_\chi^2), \quad \sigma_\chi^2 = (1 - \lambda^2) \sigma_\theta^2 \quad (5)$$

where  $\lambda$  measures *continuity*, capturing the extent to which internal successors carry forward the strategy of the regime that produced them, and where  $\chi_{i'}$  captures individual heterogeneity in executive ability not inherited from the predecessor, which has implied stationary variance  $\sigma_\chi^2$ . We parameterize  $\lambda$  as a fraction of  $\phi$  in (4),  $\lambda = \kappa\phi$  with  $\kappa \in [0, 1]$ , so that a successor inherits at most as much predecessor productivity as the predecessor herself would carry into the next period.<sup>4</sup>

**Information structure.** Boards observe profitability and know its permanent component  $\iota_j$  in (1) and the population distributions of CEO productivity for internal and external talent. However, they do not observe separately the idiosyncratic shocks to profitability  $\varepsilon_{jt}$  or the contributions to profitability coming from its CEO's productivity  $\theta_{it}$  and from the industry factor  $g_{nt}$ . Instead, the board forms beliefs about the evolving latent quantities  $\theta_{it}$  and  $g_{nt}$  using two signals. First, realizations of time-varying profitability, as governed by (2), generate noisy signals  $u_{jt}$  of CEO productivity:

$$u_{jt} \equiv y_{jt} - \rho y_{jt-1} = (1 - \rho) \theta_{it} + g_{nt} + \varepsilon_{jt} \quad (6)$$

which confound the CEO's contribution with industry conditions and idiosyncratic shocks. Second, the board observes a noisy signal  $s_{nt}^g$  of the industry factor itself:

$$s_{nt}^g = g_{nt} + \omega_{jt}, \quad \omega_{jt} \sim N(0, \sigma_\omega^2) \quad (7)$$

reflecting the information about industry conditions available to the board from peer performance, analyst commentary, and its own reading of the environment. The precision of this signal determines how well the board can separate industry luck from CEO productivity. This information structure with latent CEO ability and industry factor is motivated by a substantial body of evidence showing that industry conditions leak into decisions which, under the informativeness principle of Holmström (1979), should be purged of them.<sup>5</sup> Rather than

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<sup>3</sup>Formally, we take  $(\theta_{it}, \theta_{i't})$  to be jointly normal with  $\theta_{it} \sim N(\mu_{h_i}, \sigma_\theta^2)$ ,  $\theta_{i't} \sim N(\mu_I, \sigma_\theta^2)$ , and  $\text{Corr}(\theta_{it}, \theta_{i't}) = \lambda$ . Conditioning on  $\theta_{it}$  delivers a normal distribution with mean  $\mu_I + \lambda(\theta_{it} - \mu_{h_i})$  and variance  $(1 - \lambda^2)\sigma_\theta^2$ .

<sup>4</sup>The restriction  $\lambda \leq \phi$  rules out the pathological region in which the board would prefer to fire a well-performing incumbent in order to “harvest” a superior copy of her.

<sup>5</sup>CEOs are more likely to be dismissed after industry downturns (Jenter and Kanaan, 2015; Kaplan and Minton, 2012) and are rewarded for sector luck in pay (Bertrand and Mullainathan, 2001).

imposing either complete or absent industry benchmarking, we let the benchmark precision  $\sigma_\omega$  determine how much industry luck contaminates the board's assessment of its CEO.

Given the two signals, boards use Bayes' rule to update their beliefs. The pair  $(\theta_{it}, g_{nt})$  and the signals  $(u_{jt}, s_{nt}^g)$  form a linear-Gaussian filtering problem in two latent states, and beliefs are jointly normal throughout. We restrict attention to steady-state learning, so that the posterior covariance is constant at its stationary value and the board's beliefs about its CEO are summarized by the posterior mean  $\tilde{\theta}_{it}$ . Under this restriction the board's uncertainty about its CEO is at its stationary level at every tenure, and the updating gain characterized in Proposition 1 is constant over a spell.<sup>6</sup> Proposition 1 characterizes the resulting belief dynamics; all of the learning mechanics used in the remainder of the paper follow from it.

**Proposition 1** (Belief dynamics). *Under the information structure (6)-(7) and steady-state learning, the board's belief about its incumbent CEO evolves according to*

$$\tilde{\theta}_{it+1} = \mu_h + \phi(\tilde{\theta}_{it} - \mu_h) + \phi K_u \left[ \underbrace{(u_{jt} - \hat{u}_{jt})}_{\text{performance surprise}} - \psi_g \underbrace{(s_{nt}^g - \tilde{g}_{nt})}_{\text{benchmark surprise}} \right] \quad (8)$$

where  $\hat{u}_{jt} = (1 - \rho)\tilde{\theta}_{it} + \tilde{g}_{nt}$  is the predicted profitability innovation,  $\tilde{g}_{nt}$  is the board's belief about the industry factor, and the updating gain  $K_u$  and the benchmark-adjustment share  $\psi_g$  are constants determined by the stationary filtering problem. Moreover, if industry persistence is lower than executive productivity persistence ( $\phi_g \leq \phi$ ):

- (i)  $K_u > 0$  and  $\psi_g > 0$ : beliefs rise with unexpectedly good firm performance, and fall when the industry benchmark is unexpectedly favorable;
- (ii)  $\psi_g \rightarrow 1$  as  $\sigma_\omega \rightarrow 0$  and  $\psi_g \rightarrow 0$  as  $\sigma_\omega \rightarrow \infty$ .

PROOF: See Appendix B.

The updating rule in Proposition 1 indicates that the board evaluates its CEO on *benchmark-adjusted* performance. It nets out a fraction  $\psi_g$  of the industry surprise before crediting the CEO, and  $K_u$  governs the overall speed at which beliefs respond to news. Part (i) formalizes how the two signals jointly identify the CEO's contribution: holding performance fixed, good news about the industry is bad news about the CEO. Part (ii) pins

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<sup>6</sup>This restriction is standard in models of learning about a time-varying latent state; see Demarzo and Sannikov (2017) for a prominent example. It concerns the board's beliefs about the executive it currently employs: its uncertainty about a *candidate* at the moment of appointment is a separate object and does differ by type, since inheritance and the screening signal both reduce it for an internal candidate. In Appendix F we relax this restriction and allow non-stationary learning; the estimates barely move. We therefore adopt stationary learning as our benchmark for analytic and computational convenience.

the degree of industry-adjustment as a function of benchmark noise  $\sigma_\omega$ . When  $\psi_g = 1$ , the board nets out the entire industry surprise: relative performance evaluation (RPE). When  $\psi_g = 0$ , industry luck is fully credited to the CEO: absolute performance evaluation (APE). Intermediate values capture partial adjustment: as the benchmark ranges from perfectly revealing to uninformative,  $\psi_g$  falls from one to zero.<sup>7</sup> Given the remaining parameters, every value of  $\psi_g \in (0, 1)$  maps one-to-one from benchmark noise  $\sigma_\omega$ ; we therefore parametrize the attribution problem and directly estimate  $\psi_g$ , which is more economically interpretable than  $\sigma_\omega$ .

**Screening.** Boards have prior information about internal talent but they have no additional information about external candidates prior to hiring. Concretely, before choosing between its two candidates, the board observes a signal  $s_{it}$  about the internal candidate:

$$s_{it} = \chi_{i'} + \nu_{it}, \quad \nu_{it} \sim N(0, \sigma_\nu^2) \quad (9)$$

centered on  $\chi_{i'}$ , the candidate's heterogeneity in productivity in (5) that is not already predictable from the predecessor. Using Bayes' rule, the signal shifts the board's belief about the insider candidate by:

$$\xi_{it} = \mathbb{E}[\chi_{i'} \mid s_{it}] = \psi_s s_{it}, \quad \psi_s = \frac{\sigma_\chi^2}{\sigma_\chi^2 + \sigma_\nu^2}, \quad \xi_{it} \sim N(0, \psi_s \sigma_\chi^2). \quad (10)$$

No analogous signal is available for the outsider. The signal variance  $\sigma_\nu^2$  measures the precision of boards' screening of internal candidates: the gap between the prior information boards possess about insider and outsider candidates captures the extent to which fitness for the CEO position is revealed through experience in lower ranks within the firm. The parameter  $\psi_s$  therefore summarises everything the firm has learned about an internal candidate before the appointment, whether through direct assessment or through the promotion decisions that produced the candidate. Screening operates through selection: because the board acts on  $\xi_{it}$ , sharper screening (smaller  $\sigma_\nu$ ) raises the average productivity and reduces the dispersion of the insiders actually appointed, relative to the pool of internal candidates. In parallel with our treatment of the attribution parameter  $\psi_g$  in (8), we parametrize screen-

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<sup>7</sup>The interpretation of  $\psi_g$  as an informational share can be made exact in a one-shot version of the learning problem. With a single performance observation  $u = (1 - \rho)\theta + g + \varepsilon$  and benchmark signal  $s^g = g + \omega$ , the posterior mean of  $\theta$  loads on  $u$  and  $s^g$  with coefficients whose ratio implies  $\psi_g = p_g / (p_g + \sigma_\omega^2)$  exactly, where  $p_g$  denotes the prior variance of the industry factor: the share of industry uncertainty that the benchmark resolves, the precise analogue of  $\psi_s$  in the screening problem (Equation (10)). In the fully dynamic problem,  $\psi_g$  retains this essence, with prior variances replaced by their steady-state counterparts and a more involved functional form.

ing directly by  $\psi_s \in [0, 1]$ : the share of the candidate-specific uncertainty  $\sigma_\chi^2$  that the board resolves before hiring, with the signal noise  $\sigma_\nu$  implied.<sup>8</sup> Because the board does not observe the productivity of the departing CEO, only having a posterior belief  $\tilde{\theta}_{it}$  about it, the board's beliefs about the incoming internal candidate's productivity apply the inheritance mapping to the posterior mean  $\tilde{\theta}_{it}$ . Hence, since the board has rational expectations, its beliefs about outsider and insider candidates upon turnover are:

$$\tilde{\theta}_0^O = 0, \quad \tilde{\theta}_0^I(\xi) = \mu_I + \lambda(\tilde{\theta}_{it} - \mu_{h_i}) + \xi_{it}, \quad (11)$$

so that a predecessor perceived as strong raises the board's assessment of the internal bench, and the realized signal moves its assessment of the particular candidate at hand. These entry conditions are where the board's information advantage about insiders resides: its belief about an insider impounds real information about the candidate, the predecessor's assessed trajectory and the screening signal, while its belief about an outsider is the population mean and contains no candidate-specific information. This asymmetry in pre-hire information is in line with a long-standing notion in personnel economics: firms know more about the workers they already employ than about outside applicants (Lazear, 1998; DeVaro and Morita, 2013), and boards assess executive talent by observing it in lower ranks (Hermalin and Weisbach, 2017; Ferreira and Nikolowa, 2023).

### 3.1 The Board's Problem

The equity value of the firm is given by:

$$P_{jt} = \mathbb{E}_t \left[ \sum_{s=0}^{\infty} \beta^s b_{jt+s} Y_{jt+s} \right]. \quad (12)$$

The flow term  $b_{jt} Y_{jt}$  represents firm profits, obtained by multiplying firm assets  $b_{jt}$  by profitability  $Y_{jt}$ , and  $\beta \in (0, 1)$  is the discount factor. We assume firm assets are constant for simplicity, thus  $b_{jt} \equiv b_j$  for all  $t$  and assets drop from boards' optimization problem. Were the board to perfectly advance the interests of shareholders, they would choose personnel management policies to maximize (12). However, we allow for the possibility that board decision-making is swayed by their private interests. In particular, we assume the board's

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<sup>8</sup>The assumption is that boards know more about internal candidates than external ones, not that they know nothing about the latter. Outside candidates are screened in practice, and Fee and Hadlock (2003) show that executives moving into CEO positions are selected on their prior employer's performance record. Because  $\psi_s$  is identified from the difference in residual performance variation across insider and outsider spells, it measures the gap in pre-hire information between the two rather than its absolute level for either type.

flow utility is given by:

$$U_{jt} = Y_{jt} - \pi d_{jt}^f + \gamma (d_{jt}^f + v_{jt}) \mathbf{1}[h_{it} = I] \quad (13)$$

where the two non-pecuniary terms reflect the wedge separating the preferences of shareholders and the board. The board incurs a non-pecuniary cost  $\pi$  from firing its CEO and it has a non-pecuniary preference  $\gamma$  for hiring a replacement from inside the firm.<sup>9</sup> These features of the model allow us to assess the relative importance of non-pecuniary motives for promoting from within the firm versus profit-driven incentives. The two wedges correspond to distinct governance frictions:  $\pi$  reflects CEO entrenchment that depresses turnover below the shareholder optimum (Hermalin and Weisbach, 1998; Taylor, 2010), while  $\gamma$  nests the view that boards favor insiders, or handicap outsiders, for reasons unrelated to value (Agrawal et al., 2006; Mobbs and Raheja, 2012).

The board chooses firing and hiring policies to maximize the expected present value of its flow utility:

$$\max \mathbb{E}_t \left[ \sum_{s=0}^{\infty} \beta^s U_{jt+s} \right] \quad (14)$$

which differs from shareholder value (12) only through the non-pecuniary terms: when  $\pi = \gamma = 0$  the two objectives coincide, up to the constant scale  $b_j$ , and the board's policy maximizes firm value. The remainder of this section represents (14) recursively.

**Timing.** Events within period  $t$  unfold in five steps. First, the retirement shock is realized: the incumbent departs voluntarily with exogenous probability  $\delta(\tau_{it})$ , a function of tenure that we estimate in a first stage.<sup>10</sup> Second, if the incumbent does not retire, the board has the alternative to retain or fire them (incurring the non-pecuniary cost  $\pi$ ). Third, if the position becomes vacant, whether by retirement or dismissal, the board draws one candidate from each talent pool, internal and external, observes the screening signal  $s_{it}$  about the internal candidate, and appoints either the insider (enjoying  $\gamma$ ) or the outsider, depending on its initial beliefs. Fourth, profitability  $Y_{jt}$  is realized under the incumbent or the newly appointed CEO, net of the turnover cost  $c$  if a succession occurred. Lastly, the board observes the profitability innovation  $u_{jt}$  and the benchmark signal  $s_{nt}^g$  and updates its beliefs according to Proposition 1, delivering the state on which next period's decisions are based. The decision to retain or fire is therefore made under beliefs that reflect performance through  $t - 1$ .

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<sup>9</sup>We assume the non-pecuniary cost of firing a CEO remain stable over time. This is consistent with evidence showing that board composition is highly persistent (Graham et al., 2020).

<sup>10</sup>In practice  $\delta(\tau)$  is logistic in a cubic in tenure; Appendix D gives the specification, the estimates, and the implied hazard profile.

**The value of retention and turnover.** Given state  $x_{jt} = (\tilde{\theta}_{it}, \tilde{g}_{nt}, y_{jt-1}, \tau_{it}, h_i)$ , which includes the board's beliefs about the incumbent CEO and the industry factor, lagged performance, and the tenure and type of the incumbent CEO, the board's value conditional on retaining the CEO is given by (suppressing the firm subscript):

$$V_0(x_t; h) = \mathbb{E} \left[ Y_t + \beta \left( v_{t+1} R(x_{t+1}) + (1 - v_{t+1}) V(x_{t+1}; h) \right) \right] \quad (15)$$

where  $v_{t+1}$  is the indicator of whether the current CEO will depart voluntarily at the beginning of next period. The replacement value  $R(x_t)$  and the value of having the option to fire  $V(x_t; h)$  are respectively given by:

$$R(x_t) = \mathbb{E}_\xi \left[ \max \{ V_0(x_t^I(\xi); I) + \gamma, V_0(x_t^O; O) \} \right] \quad (16)$$

$$V(x_t; h) = \max \{ V_0(x_t; h), -\pi + R(x_t) \} \quad (17)$$

where  $x_t^I(\xi)$  and  $x_t^O$  denote the entry states of the insider and outsider candidates, with beliefs given by (11) and tenure reset to one. The replacement value  $R(x_t)$  reflects the board's optimal hiring choice between internal and external talent following a turnover event, the function  $V(x_t; h)$  is the value of having the option to trigger such turnover. Note that the monetary impact of forced turnover (the cost  $c$ ) is implicit in the conditional value functions in (16), as they incorporate the impact of CEO departure on profits. The optimal firing and hiring decisions characterized in this section capture the effect of prior performance on current beliefs, including the extent to which industry luck is attributed to the CEO and the expectations boards have regarding future firm performance. These expectations are affected by the persistence of CEO and industry productivity, by the persistence in profitability itself, by management continuity across successions, and by boards' screening of internal talent.

Finally, equity value is the expected present value (12) of shareholder cash flows evaluated under the board's policy: it incorporates the monetary cost  $c$  of succession but excludes the non-pecuniary components  $\pi$  and  $\gamma$ , which enter board utility rather than cash flow. Abnormal returns in the model are constructed from  $P_t$  exactly as in the data, as returns net of the industry benchmark. Because the price capitalizes the board's beliefs, abnormal returns move with the *revisions* of those beliefs: news moves the price exactly insofar as it changes the board's assessment of the firm's leadership. Abnormal returns are therefore the model's natural summary of the arrival of decision-relevant information and we accordingly use returns as the performance measure for the hiring and firing margins in the estimation process described below.<sup>11</sup>

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<sup>11</sup>This link between belief revisions and prices is the mechanism documented empirically by Pan et al.

## 4 Identification and Estimation

We recover parameter estimates using a two-step estimation procedure. In a first-stage, we estimate the parameters of the voluntary separation process, identified off of variation in voluntary turnover rates across CEO tenure. These estimates are taken as an input in the second stage, where the remaining 15 parameters are estimated using the Simulated Method of Moments. The discount factor is calibrated throughout to  $\beta = .95$ .<sup>12</sup>

We first discuss how we disentangle the parameters governing firm profitability in (1): the permanent firm component  $(\bar{\iota}, \sigma_{\iota})$ , the idiosyncratic shock  $\sigma_{\epsilon}$ , firm persistence  $\rho$ , and the industry factor  $(\phi_g, \sigma_{\zeta})$ . The mean and the across-firm variance of profitability pin down  $\bar{\iota}$  and  $\sigma_{\iota}$ . We discipline the industry factor directly by targeting the variance and the first- and second-order autocorrelations of industry-level profitability, computed over the universe of Compustat firms within each industry. Identification of  $\rho$  and  $\sigma_{\epsilon}$  comes from the dynamics of performance within a CEO's spell. Performance moves within a spell both because of idiosyncratic shocks and because the quality of the firm's leadership evolves, and the two are distinguished by how long they last: an idiosyncratic shock feeds into subsequent profitability at rate  $\rho$  and then dies out, while a movement in quality persists at rate  $\phi$  and continues to affect performance for the remainder of the spell. We target the within-spell variance of industry-adjusted profitability together with the variances of its one-year and two-year changes. Differencing removes the permanent firm component, and comparing the change variances across the two horizons measures how quickly the transitory component decays: change variances that grow little between the one- and two-year horizons indicate rapid mean reversion, while variances that continue to grow indicate longer-lived innovations. Together with the pooled autocorrelations of profitability, these moments are primarily informative about  $\rho$  and  $\sigma_{\epsilon}$ .

The tenure profile of performance differences across CEO types carries information about the succession channels: inheritance, screening, and the type mean. To separate these channels, we target the coefficients of the following regression of profitability on tenure-binned type indicators:

$$y_{ijt} = \sum_b \delta_b^y \mathbf{1}[\tau_{ijt} \in b] \cdot \text{insider}_{ijt} + \sum_{b \geq 2} \delta_b^0 \mathbf{1}[\tau_{ijt} \in b] + \delta_0^y \mathbf{1}[\text{turnover}]_{jt} + \alpha_j + \alpha_t + \epsilon_{ijt}^y \quad (18)$$

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(2015), who show that uncertainty about CEO ability is capitalized into stock prices and resolves gradually over tenure, and it is consistent with the long-standing evidence that prices respond to top-management changes (Warner et al., 1988).

<sup>12</sup>Appendix C lists the targeted moments in full and gives details on the weight matrix, the numerical routine, the standard-error formulas, and model fit.

where the tenure bins  $b$  are  $\{1, 2, 3, 4-6, 7-9, 10+\}$  and  $\alpha_j$  and  $\alpha_t$  are firm and year effects. The tenure-bin main effects control for the common tenure profile of performance (bin 1 is normalized to zero, absorbed by the fixed effects), so each  $\delta_b^y$  measures the insider-outsider performance difference within tenure bin  $b$ . The tenure profile of the coefficients  $\delta_b^y$  traces the life cycle of the insider premium, and its shape carries information about distinct channels. The initial level of the premium reflects the informational advantages insiders enjoy at the time of hire, to which inheritance, screening, and the type mean all contribute; its long-tenure level carries information about  $\mu_I$ , mediated by survival selection. The profile's erosion over tenure reflects the equilibrium catch-up of surviving outsiders as boards learn and replace poor performers, and so disciplines the speed of learning together with the firing margin. The coefficient on the turnover indicator, capturing the dip in performance around CEO transitions, is most informative about the turnover cost  $c$ .

The dispersion of CEO quality is identified from the between-CEO variation in average within-firm performance among outsider spells: outsider appointees are unscreened draws from the population, so the spread of their spell-level performance, after removing firm effects, is informative about  $\sigma_\theta$ ; these are the between-CEO differences in performance that Bertrand and Schoar (2003) attribute to managers, computed here for the unscreened CEO type. Since  $\sigma_\theta^2 = \sigma_\eta^2 / (1 - \phi^2)$ , this moment together with the persistence  $\phi$  determines the estimated innovation variance  $\sigma_\eta$ .

The precision of boards' internal information,  $\psi_s$ , is identified from the composition of insider hires. As the precision of the signal increases, board selectivity over insider quality increases, which concentrates the distribution of realized insider productivity among higher values. Thus, since firms only receive prior signals about the quality of insider candidates, the difference in residual performance variation across insider and outsider CEO spells identifies  $\psi_s$ : we target the between-CEO variance of average within-firm performance separately by type, along with the share of successions filled internally. The choice margins provide additional discipline: as screening precision falls, the insider share of successions rises and the forced-turnover rate falls. Both follow from the board being unable to tell strong internal candidates from weak ones: it promotes insiders more indiscriminately, so the share rises, and it dismisses incumbents less often, because no successor can be identified as clearly better. As a result, the pair of margins over-identifies the screening precision. Continuity ( $\lambda$ ) ties the entering productivity of internal successors to the productivity of the regime that produced them. The parameter  $\lambda$  is therefore disciplined by the dependence of hiring choices on the predecessor's track record.

The remaining parameters, the persistence of quality  $\phi$ , the attribution parameter  $\psi_g$ , and the preference parameters  $\pi$  and  $\gamma$ , are identified off of the hiring and firing margins.

We target the coefficients of regressions of the firing and hiring decisions on the information available to the board at the time of the decision:

$$d_{ijt}^f = \delta_r^d R_{ijt}^{\text{rec}} + \delta_o^d R_{ijt}^{\text{old}} + \delta_g^d R_{jt}^{\text{ind}} + \alpha_j + \alpha_t + \varepsilon_{ijt}^d \quad (19)$$

$$\text{insider}_{i'jt} = \delta_r^h R_{ijt}^{\text{rec}} + \delta_o^h R_{ijt}^{\text{old}} + \delta_g^h R_{jt}^{\text{ind}} + \alpha_j + \alpha_t + \varepsilon_{ijt}^h \quad (20)$$

where  $R_{ijt}^{\text{rec}}$  and  $R_{ijt}^{\text{old}}$  are the firm's recent and longer-lagged market-adjusted returns under the current (in (19)) or departing (in (20)) CEO,  $R_{jt}^{\text{ind}}$  is the contemporaneous industry return, and  $i'$  indexes the incoming CEO.<sup>13</sup> The performance coefficients discipline the mapping from information to decisions: the relative weight on recent versus longer-lagged performance reflects the speed at which boards update, which the learning parameters must reproduce. The same recency structure is what identifies the persistence of quality: the more persistent is quality, the more weight older performance carries relative to recent performance in the board's firing decisions (Jenter and Lewellen, 2020). Returns enter the regressions decomposed into their firm-specific and industry components, so the industry coefficients measure the extent to which industry conditions enter personnel decisions holding the firm's own performance fixed. Industry conditions reach the board's decisions only through its beliefs about the CEO, so a board that fully filters industry performance from its assessments ( $\psi_g = 1$ ) fires and hires without regard to the industry component of returns, while a board that credits industry performance to the CEO responds to it as it does to firm-specific news; the response of turnover to industry returns relative to firm-specific returns therefore pins down  $\psi_g$ . Conditional on the parameters governing profitability and belief formation, and in particular on the persistence of quality,  $\pi$  is identified off of the empirical rate of forced CEO termination, and  $\gamma$  off of the level of the insider share.<sup>14</sup> Screening also moves the insider share, but in the opposite direction: sharper screening makes boards more selective and *lowers* the share, because boards decline internal candidates they would otherwise have promoted, whereas a stronger preference for insiders raises it. The level of the share is therefore jointly determined by the two, and they are separated by what each implies for the quality of the insiders appointed. Sharper screening raises the average productivity of appointed insiders and compresses their dispersion. A preference for insiders promotes candidates the

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<sup>13</sup>Appendix C gives the exact windows and variable definitions. Both regressions are estimated on the same samples, with the same fixed effects, in the model and in the data. The firing regression additionally includes an insider indicator as a control, so that the performance coefficients are identified within CEO type; its coefficient is not among the targeted moments.

<sup>14</sup>Jenter and Lewellen (2020) caution that some nominally voluntary exits are in fact performance-induced. To the extent such departures remain in our voluntary margin, the forced-turnover rate understates the true incidence of belief-driven dismissal and the estimated firing margin understates the responsiveness of turnover to beliefs, making our estimates of what boards know conservative.

board would otherwise pass over, which lowers their average productivity and widens their dispersion. Because hiring and forced turnover are endogenous, the identification of these parameters critically depends on our modeling of the hiring and firing decisions, which takes into account time-varying unobservable beliefs and firm performance persistence. Finally, since the hiring gradient in (20) also carries information about the degree of screening attribution ( $\psi_s$ ) and the continuity in productivity associated with internal promotions ( $\lambda$ ), the variance moments by CEO type are crucial to separate how much variation in the gradient corresponds to preferences, how much to screening, and how much to continuity.

Table 4: Model Fit

| Moment                                      | Model  | Data   | $t$   | Moment                                     | Model  | Data   | $t$   |
|---------------------------------------------|--------|--------|-------|--------------------------------------------|--------|--------|-------|
| <i>Profitability dynamics and variances</i> |        |        |       | <i>Hiring</i>                              |        |        |       |
| ROA AR(1)                                   | 0.709  | 0.649  | 2.40  | recent performance                         | 0.019  | 0.058  | -2.98 |
| ROA AR(2)                                   | 0.150  | 0.168  | -0.75 | older performance                          | 0.024  | 0.029  | -0.39 |
| Mean ROA                                    | 0.120  | 0.120  | 0.00  | industry performance                       | 0.005  | 0.001  | 0.24  |
| Var. firm effect                            | 0.0106 | 0.0108 | -0.24 | insider share                              | 0.658  | 0.662  | -0.41 |
| Within-spell var.                           | 0.0037 | 0.0036 | 0.41  | <i>Firing</i>                              |        |        |       |
| Var. 1-yr change                            | 0.0050 | 0.0048 | 0.83  | recent performance                         | -0.011 | -0.011 | -0.25 |
| Var. 2-yr change                            | 0.0074 | 0.0072 | 0.46  | older performance                          | -0.009 | -0.007 | -1.25 |
| Btw-CEO var., ins.                          | 0.0022 | 0.0023 | -0.92 | industry performance                       | -0.004 | -0.004 | -0.36 |
| Btw-CEO var., out.                          | 0.0026 | 0.0038 | -3.17 | forced rate                                | 0.027  | 0.028  | -0.56 |
| <i>Insider premium by tenure</i>            |        |        |       | <i>Industry factor</i>                     |        |        |       |
| Year 1                                      | 0.018  | 0.022  | -0.86 | $g$ AR(1)                                  | 0.714  | 0.710  | 0.07  |
| Year 2                                      | 0.015  | 0.011  | 1.12  | $g$ AR(2)                                  | 0.416  | 0.401  | 0.22  |
| Year 3                                      | 0.006  | 0.013  | -1.81 | Var. industry mean                         | 0.0007 | 0.0011 | -1.39 |
| Years 4-6                                   | 0.002  | 0.003  | -0.25 | <i>Baseline tenure profile (outsiders)</i> |        |        |       |
| Years 7-9                                   | 0.001  | 0.006  | -1.26 | Year 2                                     | -0.001 | 0.006  | -1.72 |
| Years 10+                                   | 0.002  | 0.006  | -1.00 | Year 3                                     | 0.009  | 0.003  | 1.37  |
| Succession dip                              | -0.007 | -0.005 | -0.70 | Years 4-6                                  | 0.015  | 0.013  | 0.43  |
|                                             |        |        |       | Years 7-9                                  | 0.016  | 0.012  | 0.65  |
|                                             |        |        |       | Years 10+                                  | 0.014  | 0.015  | -0.19 |

Notes: Model moments are computed at the estimated parameter vector;  $t$ -statistics are the difference between model and data moments divided by the empirical standard error of the moment. The baseline tenure-profile rows are the tenure-bin main effects of the premium regression (year 1 is the base), which pin the performance profile under outsider CEOs.

We target a total of 32 moments to recover 15 model parameters. A summary of model fit is presented in Table 4. The estimated model fits the data well. Table 4 compares all 32 targeted moments with their model counterparts: 29 of 32 are within two standard errors of their empirical values. The largest discrepancy is the between-CEO variance among outsider spells, which the model under-produces: outsider CEOs in the data are an even more

dispersed pool than the model generates. The miss is conservative for our purposes, as it implies that the informational gap between internal and external candidates is, if anything, larger than the model’s estimate of it. We cluster standard errors by firm and report them alongside the estimates in Table 5. In Appendix E, we formalize the mapping between moments and parameters using the sensitivity matrix outlined in Andrews et al. (2017), which confirms that each estimate is driven primarily by the moments highlighted above.

## 5 Structural Estimates

Despite the statistically significant difference in the performance of newly appointed insiders and outsiders shown in Figure 1, a pattern at the center of an open and longstanding debate over the firm-performance consequences of internal appointments (Shen and Cannella, 2002; Zhang and Rajagopalan, 2010; Quigley et al., 2019), our estimates in Table 5 reveal only a slight difference of  $\mu_I = 0.61\%$  of firm assets in the long-run productivity of internal and external candidates.<sup>15</sup> This is a modest gap relative to the estimated cross-sectional stationary dispersion of CEO productivity,  $\sigma_\theta = \sigma_\eta / \sqrt{(1 - \phi^2)} \approx 3.5\%$  of assets, which echoes the descriptive finding of Cziraki and Jenter (2021) that internal promotion dominates the market for CEOs even as insiders are not intrinsically superior. The discrepancy is driven in good part by what boards *know* about their candidates, rather than by what the candidates *are* on average.

Three main mechanisms explain the discrepancy, and they underscore the importance of accounting for the endogeneity of hiring and retention. First, internal successors inherit part of the trajectory of the regime that produced them ( $\lambda = 0.35$ ), rendering them more attractive candidates during good times, and resulting in genuinely better fundamentals from the start. This is consistent with prior descriptive observations showing that insiders are disproportionately likely to be hired by firms that have been performing well, while boards replacing failing CEOs tend to look outside (Parrino, 1997; Huson et al., 2001), an empirical regularity our model reproduces. Second, the inheritance mechanism is exacerbated by firm persistence. Given the estimated operational persistence,  $\rho = 0.49$ , firm performance following the appointment of an insider, who likely inherited better fundamentals, will on average be persistently higher. Third, and most importantly, we find that insider candidates are substantially better screened than outsiders, which likely reflects boards’ prior information on internal personnel accumulated during their tenure in junior roles before CEO appointment (Baker et al., 1994).

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<sup>15</sup>Since our measure of profitability is the industry-adjusted ROA, the parameters in Table 5 are expressed in terms of a percentage of firm assets. We relegate estimates of the voluntary turnover model to Appendix D.

Table 5: Structural Estimates

| Parameter                              | Description                                                   | Estimate       |
|----------------------------------------|---------------------------------------------------------------|----------------|
| <i>CEO productivity and succession</i> |                                                               |                |
| $\phi$                                 | Productivity persistence                                      | 0.980 (0.003)  |
| $\sigma_\eta$                          | Productivity-innovation s.d.                                  | 0.007 (0.001)  |
| $\mu_I$                                | Insider productivity premium                                  | 0.006 (0.002)  |
| $\kappa$                               | Continuity (as fraction of $\phi$ )                           | 0.358 (0.083)  |
| $\lambda = \kappa\phi$                 | Implied inheritance                                           | 0.351 (0.081)  |
| <i>Board information</i>               |                                                               |                |
| $\psi_s$                               | Screening precision (share of candidate uncertainty resolved) | 0.319 (0.076)  |
| $\sigma_\nu$                           | Implied signal noise                                          | 0.047          |
| $\psi_g$                               | Industry-performance filtering                                | 0.649 (0.192)  |
| $\sigma_\omega$                        | Implied benchmark noise                                       | 0.017          |
| <i>Firm profitability</i>              |                                                               |                |
| $\bar{\iota}$                          | Firm-effect mean                                              | 0.102 (0.004)  |
| $\sigma_\iota$                         | Firm-effect s.d.                                              | 0.100 (0.004)  |
| $\rho$                                 | Operational persistence                                       | 0.493 (0.038)  |
| $\sigma_\varepsilon$                   | Transitory shock s.d.                                         | 0.061 (0.002)  |
| $\phi_g$                               | Industry-factor persistence                                   | 0.442 (0.081)  |
| $\sigma_\zeta$                         | Industry-factor innovation s.d.                               | 0.017 (0.003)  |
| $c$                                    | Monetary turnover cost                                        | 0.0003 (0.003) |
| <i>Board preferences</i>               |                                                               |                |
| $\pi$                                  | Non-pecuniary firing cost                                     | 0.077 (0.020)  |
| $\gamma$                               | Insider hiring preference                                     | 0.004 (0.006)  |

Notes: Standard errors are shown in parentheses. They are computed from a robust sandwich formula that adjusts for simulation error, with the moment covariance matrix bootstrapped and clustered by firm (by industry for the industry moments). Since our measure of profitability is the industry-adjusted ROA, the parameters are expressed in terms of a percentage of firm assets. The standard errors for  $\psi_s$  and  $\lambda$  are computed by the delta method from the full estimated parameter covariance.

Our estimates describe what boards know at each of their two margins of action: what they learn about a candidate before an appointment, which we refer to as *selection*, and what they learn about an incumbent executive as they decide whether to retain them, which we refer to as *monitoring*. Consider first the selection margin. Boards bring two kinds of information to an internal appointment that they lack for an external one: a pre-hire signal about the insider candidate capturing their prior knowledge about the prospect, and what they have already learned about the outgoing CEO whose quality the internal prospect will partially inherit. The estimated precision of the pre-hire signal ( $\psi_s = 0.32$ ) indicates that boards resolve roughly a third of the uncertainty about an insider candidate’s productivity innovation ( $\chi_{i'}$  in (5)) before making the hire, and the confidence interval rejects both no

screening and perfect screening.<sup>16</sup> The estimated value of  $\lambda$  implies that inheritance resolves about six percent of the variance in an internal candidate’s productivity, against twenty-eight percent for the signal.<sup>17</sup> Screening individual candidates, rather than knowledge about their inherited path, is thus the more important source of discriminating information.

Panel (a) of Figure 2 shows the impact of screening on the distribution of productivity among appointed CEOs. Relative to the pool of internal candidates, the productivity distribution of appointed insiders shifts to the right by 0.32 standard deviations of CEO productivity. Panels (b) and (c) show the two channels behind that shift: prior information and inheritance. Panel (b) shows that the appointment probability of an internal candidate rises from 0.18 for executives at the bottom of the productivity distribution to 0.97 for those at the top, revealing the power of screening internal prospects. Panel (c) shows that inheritance has a similar but weaker impact: the appointment probability of an internal candidate rises from 0.55 to 0.80 as the productivity of the departing CEO goes from the bottom to the top of the support. The gradient is flatter in panel (c) because quality transfers only partially across regimes, and because the board observes the departing executive’s contribution with error.<sup>18</sup>

Consider now the monitoring margin, which begins after selection, as boards attempt to learn about the evolving productivity of their CEOs through the record they generate. Our estimates indicate that the evolving CEO productivity in equation (4) is highly persistent ( $\phi = 0.98$ ) with modest volatility in year-to-year innovations ( $\sigma_\eta = 0.7\%$  of assets). Hence, productivity differences across CEOs ( $\sigma_\theta \approx 3.5\%$  of assets) are large and durable, and the environment resembles the fixed-type benchmark of Jovanovic (1979) and Miller (1984). The board’s problem is therefore dominated by learning about a durable attribute of considerable value, and the pace of that learning is set by the signal-to-noise ratio of profitability: each annual realization mixes the CEO’s contribution with transitory shocks roughly twice as volatile ( $\sigma_\varepsilon = 0.061$  against  $\sigma_\theta = 0.035$ ). The implied pace of learning is slow. At the stationary solution, a full year of information (the realized profitability innovation together

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<sup>16</sup>The implied signal noise is  $\sigma_\nu = 0.047$  while the standard deviation of a candidate’s productivity innovation is  $\sigma_\chi = 0.032$ . The two combine into the screening precision as  $\psi_s = \sigma_\chi^2 / (\sigma_\chi^2 + \sigma_\nu^2)$ . Hence, the noise in what a board observes about an internal candidate ( $\sigma_\nu^2$ ) is roughly twice the size of the quality differences it is trying to detect ( $\sigma_\chi^2$ ).

<sup>17</sup>Both are variance reductions relative to  $\sigma_\theta^2$ , the board’s uncertainty about an outsider candidate. A board holding posterior variance  $\Sigma$  about the departing CEO faces prior variance  $\lambda^2\Sigma + (1 - \lambda^2)\sigma_\theta^2$  about an internal candidate. Inheritance is therefore worth  $\lambda^2(\sigma_\theta^2 - \Sigma)$ , or 6.2 percent of  $\sigma_\theta^2$  at our estimates with  $\Sigma$  at its steady-state value. The signal resolves a further  $\psi_s(1 - \lambda^2)\sigma_\theta^2$ , or 27.9 percent.

<sup>18</sup>Mechanically, given that boards are assumed not to have information in excess of rational expectations for outsiders, the distributions in panel (a) would look identical for candidates and appointed outsiders and the profile in panel (b) would be flat. Moreover, since outsiders CEO productivity at firm  $j$  is not affected by  $j$ ’s prior executive, the profile in panel (c) would also be flat for outsiders.

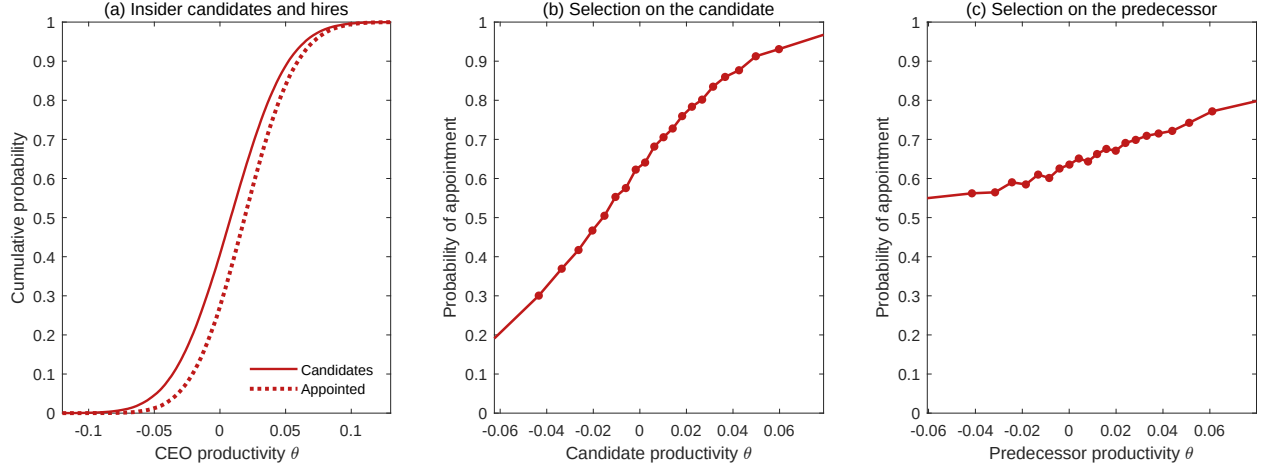


Figure 2: Selection at the Appointment Margin

Notes: Panel (a) plots CDFs of productivity for internal candidates and for the internal candidates appointed; the candidate distribution contains the internal candidate on offer at each succession, whether or not that candidate was appointed. Panel (b) plots the probability of appointment against the candidate's own productivity. Panel (c) plots the probability of appointment against the productivity of the departing CEO.

with the industry benchmark) resolves only 3.8% of the board's uncertainty about its CEO's current productivity,<sup>19</sup> a pace consistent with the gradual learning that Taylor (2013) infers from CEO pay and with the evidence of Pan et al. (2015) that uncertainty about CEO ability remains substantial years into tenure. A board that has just appointed an executive therefore learns only gradually whether it chose well, which is part of what makes the pre-appointment signal valuable.

Our estimated attribution parameter ( $\psi_g = 0.65$ ) indicates that boards filter most, though not all, industry performance from their assessments of CEO productivity, crediting the remaining 35% of an industry surprise to the CEO. Boards are thus considerably more sophisticated in their treatment of industry conditions than an absolute-performance benchmark ( $\psi_g = 0$ ) would suggest. Prior reduced-form evidence in the literature has established that industry conditions reach personnel decisions (Kaplan and Minton, 2012; Jenter and Kanaan, 2015) and pay (Bertrand and Mullainathan, 2001). Our results complement this literature by establishing how much industry performance boards credit to the CEO. Given our estimated magnitude of attribution, partial responsiveness to industry returns emerges as the rational use of an imperfect benchmark rather than a failure to benchmark at all.

Finally, we turn to the parameters describing boards' deviations from profit maximiza-

<sup>19</sup>Stationary learning holds the board's posterior variance constant across periods, not within them. Within a period the board updates on the year's signals, which lowers the variance of its belief about  $\theta_{it}$  by 3.8%, a standard deviation of 2.48% of assets falling to 2.43%; productivity then drifts and restores it before the next observation.

tion. At the hiring margin, we find minimal non-pecuniary taste for insiders ( $\gamma = .004$ ) that is statistically indistinguishable from zero. This result suggests that no such taste is necessary to account for the prevalence of internal promotion. Instead, the insider share of successions is fully rationalized by information and productivity: boards promote from within because they know more about internal candidates and because internal candidates can inherit successful regimes, not because boards favor them. According to our estimates, the handicap outsiders appear to face in successions (Agrawal et al., 2006) emerges as a consequence of asymmetric information rather than board favoritism for internal prospects, which is consistent with internal governance grooming and revealing inside talent (Mobbs and Raheja, 2012). At the firing margin, by contrast, we find that the estimated non-pecuniary firing cost ( $\pi = 0.077$ ) is material; it amounts to roughly 65% of a year of average operating profitability, in line with the substantial entrenchment frictions found in the literature (Taylor, 2010; Lyman, 2024). This wedge governs how readily boards act on what monitoring reveals, impacting the value of monitoring itself. We defer the welfare implications of these frictions, and the prices they imply for information at each margin, to the counterfactual analysis in Section 6.

## 6 The Value of Board Information

We now use the estimated model to quantify the value of board information at each of the two margins, selection and monitoring. We conduct three counterfactual exercises that alter one or more features of the board’s information structure. First, we price the board’s information at each of the two margins separately, by replacing the board’s decision at that margin with an uninformed rule. Second, we measure how far boards stand from full information about their candidates and which frictions account for the distance. Third, we focus on the selection margin and value separately each of the advantages an internal candidate brings to the firm (firm-specific human capital, strategy inheritance and prior information), assessing the implications for the insider performance premium documented in Section 2. Each counterfactual solves the board’s problem under the new environment and simulates the economy. Our welfare metric is firm value, the expected present value of shareholder cash flows under the board’s policy, which incorporates the monetary cost of succession and excludes the non-pecuniary components of board utility. We report percentage changes relative to the estimated baseline.

## 6.1 The Value of Information at Each Margin of Action

On personnel decisions, boards are graded almost entirely on the monitoring margin, and largely through one statistic: how strongly dismissal responds to performance. A weak response is ambiguous. It may mean that a board knows too little to act on or that it is unwilling to act on what it knows, and reduced-form estimates cannot distinguish between them (Engel et al., 2003). The model separates the two because the precision of what boards observe and the wedge governing how readily they act on it are distinct mechanisms in the model associated with different parameters. With this separation in hand, we price the board’s information at each margin by replacing the board’s optimal policy with an uninformed rule. At the hiring margin, the successor’s type is drawn at random rather than conditioned on the board’s information, with sampling probabilities calibrated to the observed insider share. This way, the experiment destroys the information content of the choice without changing the composition of appointments. At the firing margin, dismissal occurs at random rather than on the board’s assessment, with the hazard calibrated to reproduce the observed forced-turnover rate, so that the experiment does not confound uninformed dismissal with more or less frequent dismissal. In each case, the other decision margin and the price of the firm re-optimize under the constraint.

Table 6 reports the results, highlighting the relative importance of board’s discretion at the hiring margin, which has received much less attention in the literature.. Boards’ discretion is worth 4.6% of firm value at the hiring margin and 4.1% at the firing margin.<sup>20</sup> However, information is worth only what boards do with it. The model contains two wedges between what a board does and what would maximize firm value: a non-pecuniary cost of dismissal,  $\pi$ , and a preference for internal appointment,  $\gamma$ . A board that reads its CEO’s performance precisely but is reluctant to act on an unfavourable reading turns less of that information into firm value than a board that acts freely, and a long literature holds that boards are frequently of the first kind (Taylor, 2010). Our estimates put the dismissal wedge at  $\pi = 0.077$  and the appointment wedge  $\gamma$  at essentially zero. The prices in Panel A of Table 6 are therefore what board information is worth to shareholders when boards act under these wedges. Next, we ask what information is worth when boards are not affected by these wedges, switching off each wedge and repeating both experiments in Panels B and C, respectively. With the dismissal wedge removed, the price of selection is essentially unchanged at 4.8% while the price of monitoring rises to 10.5%. Hence, selection gains

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<sup>20</sup>Because the board observes nothing about external candidates in the baseline, these figures depend on how much boards could know about them. Appendix F.2 re-solves the counterfactuals allowing boards a pre-appointment signal about outsiders, up to the precision they hold about insiders; the value of board information rises throughout, so the figures reported here are conservative with respect to that assumption.

Table 6: The Value of Board Information at Each Margin

|                                                                 | $\Delta$ Value (%) | $\Delta$ ROA (%) | Insider share | Forced rate |
|-----------------------------------------------------------------|--------------------|------------------|---------------|-------------|
| <i>Panel A: at the estimated parameters</i>                     |                    |                  |               |             |
| Baseline                                                        | —                  | —                | 0.658         | 0.027       |
| Uninformed selection                                            | −4.64              | −5.54            | 0.663         | 0.029       |
| Uninformed monitoring                                           | −4.10              | −4.84            | 0.674         | 0.027       |
| Uninformed at both margins                                      | −9.22              | −10.98           | 0.663         | 0.027       |
| <i>Panel B: no dismissal wedge (<math>\pi = 0</math>)</i>       |                    |                  |               |             |
| Baseline                                                        | —                  | —                | 0.675         | 0.320       |
| Uninformed selection                                            | −4.79              | −5.40            | 0.675         | 0.307       |
| Uninformed monitoring                                           | −10.47             | −12.22           | 0.684         | 0.320       |
| Uninformed at both margins                                      | −15.90             | −18.77           | 0.675         | 0.320       |
| <i>Panel C: no insider preference (<math>\gamma = 0</math>)</i> |                    |                  |               |             |
| Baseline                                                        | —                  | —                | 0.647         | 0.026       |
| Uninformed selection                                            | −4.71              | −5.65            | 0.647         | 0.027       |
| Uninformed monitoring                                           | −3.94              | −4.68            | 0.665         | 0.026       |
| Uninformed at both margins                                      | −9.12              | −10.88           | 0.648         | 0.026       |

Notes: Each row replaces the board’s decision at the indicated margin with an uninformed rule and re-solves the model, with the remaining margin and the price of the firm re-optimizing. Panels B and C first re-solve the model with the indicated preference parameter set to zero and then repeat both experiments inside that economy; percentage changes are always relative to the baseline of the panel in which they appear. In every panel the random successor is drawn at the insider share prevailing in that panel’s own baseline, and the random dismissal hazard is calibrated to reproduce that baseline’s forced-turnover rate. Holding the mix fixed at the hiring margin and the rate fixed at the firing margin is what makes the two prices comparable: each experiment removes the information content of a decision without changing how often, or in what proportions, that decision goes each way.

relative strength (Panel A) in the presence of dismissal frictions and the value of monitoring is suppressed by the wedge governing whether boards act on information at the firing margin. The results in Panel C are close to null, as one would expect given that  $\gamma$  is essentially zero: removing the insider preference leaves all three prices within simulation noise of Panel A.

## 6.2 The Distance to Full Information

Now that we have measured the value of information at each decision margin, it is worth asking how much of the total value of information boards actually capture. Table 7 answers by endowing boards with perfect information at one or both margins. Perfect *selection* lets the board observe the productivity of both candidates at the moment of appointment.

Perfect *monitoring* lets it observe the productivity of the incumbent CEO at all times.<sup>21</sup> Preference parameters are held at their estimates throughout, so these experiments vary what boards know while holding fixed how readily they act on it.

Table 7: The Distance to Full Information  
*Panel A: Firm value (% change from the estimated model)*

|                     | Baseline<br>monitoring | Perfect<br>monitoring | Gain from<br>monitoring |
|---------------------|------------------------|-----------------------|-------------------------|
| Baseline selection  | —                      | +12.62                | +12.62                  |
| Perfect selection   | +17.09                 | +18.77                | +1.68                   |
| Gain from selection | +17.09                 | +6.14                 |                         |

*Panel B: Composition of turnover*

|                    | Insider share          |                       | Forced turnover rate   |                       |
|--------------------|------------------------|-----------------------|------------------------|-----------------------|
|                    | Baseline<br>monitoring | Perfect<br>monitoring | Baseline<br>monitoring | Perfect<br>monitoring |
| Baseline selection | 0.658                  | 0.545                 | 0.027                  | 0.075                 |
| Perfect selection  | 0.598                  | 0.584                 | 0.073                  | 0.077                 |

Notes: Perfect selection endows the board with direct observation of both candidates' productivity at the moment of appointment. Perfect monitoring endows it with direct observation of the incumbent's productivity, which also eliminates misattribution of industry conditions, since a board that observes productivity has no inference problem to solve. Preference parameters are held at their estimates in every regime. In Panel A the final column and the final row report the marginal effect of perfecting one margin while holding the other fixed. Panel B reports the insider share of new appointments and the forced-turnover rate in the same four regimes.

Panel A reports changes in firm value in the four resulting regimes. Full information raises firm value by 18.8%, measuring the total cost of the information frictions boards operate under. What stands out is how much of that total either margin delivers on its own: perfect selection alone is worth 17.1% and perfect monitoring alone 12.6%. The two are therefore strong substitutes. Conditional on the other margin already being perfect, perfecting selection is worth only 6.1 points and perfecting monitoring only 1.7. Hence, a board that always appoints the better candidate has little left to learn about the CEO it installed, and a board that can identify a failing CEO at once has less need to have screened

<sup>21</sup>Perfect monitoring also removes any scope for misattributing industry conditions, since a board that observes productivity directly has no inference problem left to solve. Attribution carries little value on its own in any case: perfecting it (setting  $\psi_g = 1$ ) while holding monitoring at its estimate raises firm value by 0.05%, and removing the board's filtering of industry conditions altogether (setting  $\psi_g = 0$ ) reduces firm value by 0.08%.

the candidate beforehand. Each capability substitutes for the other.

Panel B reports the impact of full information on turnover. Better information makes boards dismiss far more readily: the forced-turnover rate is 2.7% in the estimated economy and lies between 7.3% and 7.7% in every informed regime. The reason is that the value of dismissal depends on the quality of the replacement, so a board that knows it can appoint well is more willing to remove the CEO it has. The substitution in Panel A reappears at this margin, and more sharply: perfecting monitoring raises the forced rate by 4.8 percentage points when selection is at its estimated precision, but by only 0.4 points when selection is already perfect. The insider share, meanwhile, falls in every informed regime, from 0.658 to between 0.545 and 0.598: better information makes boards less insider-prone rather than more, a point we discuss further in Section 6.3.

Taken together, the two exercises in Sections 6.1 and 6.2 place boards between two benchmarks: a board that makes personnel decisions at random, and a board that observes CEO productivity perfectly. Boards are genuinely informed: their use of information at the hiring margin alone is worth 4.6% of firm value relative to uninformed hiring. But they are far from fully informed: complete information about CEO productivity would add another 19% to firm value. The two margins stand differently in this respect: boards' existing information is worth about as much at selection as at monitoring (Table 6), but the value still unrealized is larger at the selection margin (Table 7). Boards' information about CEO quality is, by these estimates, a first-order determinant of firm value.

### 6.3 Information and the Insider Premium

The exercises above vary what boards know about candidates while holding fixed what candidates are. An internal candidate, though, differs from an external one in three ways. An insider has accumulated firm-specific human capital at lower ranks, so internal candidates are on average more productive in the job than external ones ( $\mu_I$ ). An insider also carries forward the strategy of the regime that produced them, so a successful organization continues into the successor's own tenure ( $\lambda$ ). And the board has watched an insider work for years, in junior roles where performance was visible, so it knows more about that candidate than about outsiders it could recruit ( $\psi_s$ ). The first two make insiders genuinely different executives; only the third makes them better understood ones. Table 8 prices each in turn, asking how much of the internal candidate's advantage rests on what boards know rather than on what the candidate brings. The largest single lever is screening. Removing the board's pre-hire signal about internal candidates reduces firm value by 5.9%: unable to select among internal candidates, boards promote indiscriminately (the insider share rises to 0.86), and the average

productivity of appointed insiders falls by more than half. Conversely, endowing boards with a perfect pre-hire signal raises firm value by 10.6%, short of the 17.1% available under perfect selection in Table 7: the signal sharpens what a board knows about its internal candidates while leaving the outside candidate unobserved, so roughly six percentage points of the value of selection rests on knowing who one is hiring from outside. Notably, better information makes boards *more* selective, not more insider-prone: under perfect screening the insider share falls to 0.60, as boards decline weak internal candidates they would otherwise have promoted. The insider productivity premium is the second quantitatively important channel: removing it costs 4.3% of value. Continuity, by contrast, is worth little: removing inheritance costs 0.06%, because it transmits unsuccessful regimes as readily as successful ones, and the two effects ultimately offset. Eliminating the insider advantage altogether, setting insider candidates equal to outsiders in mean, inheritance, and observability, reduces firm value by 10.2%, which measures the total value of the internal succession pipeline.<sup>22</sup> Screening accounts for much of that total, so the pipeline’s value rests largely on the information it produces: the option to select on privately observed fit, the margin that distinguishes an internal promotion from an external hire (Chan, 1996; Bidwell, 2011).

A related question is what boards would lose if the composition of appointments were dictated rather than chosen. Table 9 dictates the successor’s type, leaving every parameter of the model at its estimate. Forcing all hires to be internal costs only 1.6%, while forcing all hires to be external costs 10.0%, as it abandons both the insider premium and the inheritance of successful regimes.<sup>23</sup> This figure is close to the 10.2% cost of making insiders identical to outsiders in Table 8, and the agreement is informative, because the two experiments reach it from opposite directions: one denies boards access to the internal pool while leaving its advantages intact, the other leaves access open while stripping the advantages away. In either case, the internal pipeline is worth about a tenth of firm value.

The full-information benchmark also bears directly on the interpretation of the insider advantage. Under full information the productivity gap between insider and outsider appointees narrows from 1.9 percentage points of assets to 0.5, and the insider share falls to

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<sup>22</sup>In this experiment insider and outsider candidates are drawn from identical distributions and are identically observable, so the board’s values for the two candidates coincide; the estimated (statistically zero) insider preference  $\gamma$  then tips every succession internal, and the insider share is accordingly uninformative. The value effect is unaffected: with identical candidates, firm value is invariant to which candidate is hired, and  $\gamma$  does not enter cash flows.

<sup>23</sup>The uninformed-selection row of Table 6 is this same forced rule at the observed insider share, so the experiments trace the cost of dictated composition as the forced share rises: 10.0% at no insiders, 6.1% at an even split, 4.6% at the observed share, and 1.6% at all insiders. The comparison of the last two is informative. A board deprived of its choice would rather commit to promoting internally than reproduce its own observed mix, because the outsiders it appoints in equilibrium are appointed precisely when the internal candidate is weak; appointing that same fraction at random forfeits the reason for appointing them.

Table 8: Counterfactuals: Channel Decomposition

|                                    | $\Delta$ Value (%) | Insider share | Turnover | CEO quality (%) |           |
|------------------------------------|--------------------|---------------|----------|-----------------|-----------|
|                                    |                    |               |          | Insiders        | Outsiders |
| Baseline                           | —                  | 0.658         | 0.095    | 1.9             | 0.0       |
| No screening ( $\psi_s = 0$ )      | −5.93              | 0.862         | 0.080    | 0.8             | 0.0       |
| Perfect screening ( $\psi_s = 1$ ) | +10.60             | 0.599         | 0.142    | 3.2             | 0.0       |
| No insider premium ( $\mu_I = 0$ ) | −4.31              | 0.557         | 0.087    | 1.6             | 0.0       |
| No continuity ( $\lambda = 0$ )    | −0.06              | 0.643         | 0.100    | 1.8             | 0.0       |
| Insiders equal to outsiders        | −10.18             | 1.000         | 0.083    | 0.0             | —         |

Notes: Each row re-solves the board’s problem and re-simulates the economy under the indicated change, holding all remaining parameters at their estimates.  $\Delta$  Value is the percentage change in firm value relative to the baseline, and the next two columns report the insider share of successions and the annual turnover rate under the counterfactual policy. CEO quality is the average realized productivity  $\theta$  of appointed executives *measured in the year of appointment*, in percent of assets, reported separately by type; a dash indicates that no executive of that type is appointed. Because boards observe nothing about external candidates, appointment cannot select on outsider quality, so the outsider column simply reproduces the mean of the external pool and screening shows up entirely in the insider column.

Table 9: Counterfactuals: Forced Composition of Appointments

|                           | $\Delta$ Value (%) | Insider share | Turnover | CEO quality (%) |           |
|---------------------------|--------------------|---------------|----------|-----------------|-----------|
|                           |                    |               |          | Insiders        | Outsiders |
| Optimal hiring (baseline) | —                  | 0.658         | 0.095    | 1.9             | 0.0       |
| All insiders              | −1.64              | 1.000         | 0.107    | 0.5             | —         |
| Fifty-fifty               | −6.07              | 0.500         | 0.093    | 0.5             | 0.0       |
| All outsiders             | −9.96              | 0.000         | 0.083    | —               | 0.0       |

Notes: Each row imposes the indicated composition of appointments; the firing policy re-optimizes and the information structure is unchanged. These experiments change the mix of hires rather than the information the board brings to the choice, and are therefore distinct from the uninformed-selection row of Table 6, which holds the mix at its observed value. CEO quality is the average realized productivity  $\theta$  of appointed executives, measured in the year of appointment and defined as in Table 8; a dash indicates that no executive of that type is appointed. Comparing the first two rows isolates the value of the board’s discretion: when the board chooses, the insiders it promotes average 1.9% of assets, and when internal promotion is imposed they average 0.5%, the mean of the internal pool.

0.58. When boards know all candidates equally well, most of the insider advantage disappears. What remains, half a percentage point, is about the size of the estimated productivity premium itself ( $\mu_I = 0.6\%$  of assets), so the advantage is primarily informational rather than an endowment of the internal pipeline. This offers a resolution to the inconclusive insider-versus-outsider performance debate introduced in Section 5: the two candidate pools are similar, and most of insiders' observed advantage is manufactured by what boards know at the time of the hire. It also explains why outsider appointments are the higher-variance bet emphasized by Quigley et al. (2019): outsiders are simply unscreened draws.

## 7 Conclusion

Boards turn out to be genuinely but partially informed, and informed differently at the two margins, only one of which has a standard measure behind it. Before an internal appointment they resolve roughly a third of the uncertainty about a candidate's productivity, enough to reject both the view that internal promotion is uninformed and the view that boards are fully informed about who they are promoting. After an appointment they learn slowly, resolving only a few percent of their uncertainty about a sitting executive in a typical year. And they filter most, though not all, of industry performance out of their assessments: personnel decisions respond to industry conditions at roughly a third of the rate at which they respond to comparable firm-specific news.

Boards' use of what they know is worth about four percent of firm value at each margin, so the neglected margin is as valuable as the studied one. The value not yet captured is larger at selection, and the two margins are strong substitutes: perfect information at either alone would deliver most of what full information offers. Boards' attribution of industry performance, by contrast, carries little firm value.

The estimates also settle what the prevalence of internal promotion reflects. Given the freedom to explain it as a board preference, the model does not: the taste for insiders cannot be distinguished from zero, and the insider share is accounted for by what boards know about internal candidates and by the successful regimes those candidates inherit. The performance advantage of insiders arises at appointment rather than from a superior internal pool, and better information makes boards more selective rather than more insider-prone.

Two limitations bound these conclusions. We model no compensation contract, so the board's problem here is one of information rather than of incentive provision, and the value we assign to information is the value of better personnel decisions alone. And we treat the board as a single decision maker, setting aside the composition and independence that a large governance literature connects to personnel outcomes.

A board that rarely dismisses a poor performer may be unwilling to act or unable to tell, and the turnover-performance sensitivity cannot separate the two, although they call for different remedies: one addressed to the incentives of directors, the other to the information reaching them. Our estimates say that both bind, and that the dismissal wedge is large enough to suppress a substantial part of what monitoring information would otherwise be worth to shareholders. A weak turnover-performance sensitivity is therefore not by itself evidence that a board is failing, and distinguishing the two readings requires a measure of what boards observe rather than a better measure of what they do.

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## A Data Appendix

This appendix describes the construction of the estimation sample: the data sources, the performance measures and the industry benchmark, the CEO panel and the classification of successions, and the filters that produce the final sample.

**Sources.** We combine four sources. Compustat North America provides firm fundamentals for the universe of publicly traded firms in the United States and Canada. CRSP provides monthly stock returns, which we compound to annual returns aligned with the fiscal year. ExecuComp provides executive-level records for S&P 1500 firms, from which we build the CEO panel; we merge the full executive file rather than the CEO file alone, because an executive’s non-CEO records at the firm are what identify prior experience at the time of appointment. Finally, we classify CEO departures as forced or voluntary using the CEO dismissal database of Gentry et al. (2021), which codes the circumstances of each departure of an S&P 1500 chief executive from press coverage.

**Performance measures.** Our measure of profitability is operating return on assets: operating income before depreciation scaled by total assets. Market-adjusted stock returns are compound annual firm returns net of the compound value-weighted market return over the same fiscal year. Both raw measures are winsorized at the 1st and 99th percentiles.

**The industry benchmark.** Industry conditions are measured at the level of GICS industry groups (24 groups). For profitability and for market-adjusted returns we compute industry-year means over the entire Compustat/CRSP universe, before the merge with ExecuComp restricts the panel to S&P 1500 firms, so that the benchmark for an industry does not depend on which of its members happen to appear in the estimation sample. A firm’s industry-adjusted performance is its raw performance minus this universe mean, winsorized again at the 1st and 99th percentiles. Computing the benchmark on the universe rather than within the sample matters for the industry moments targeted in estimation: sample-only industry cells are small enough that a within-sample mean would carry substantial sampling noise into the industry-adjusted series.

**The CEO panel.** From the merged Compustat-CRSP-ExecuComp panel we retain firm-years with non-missing operating income, assets, and fiscal year, and drop firms with non-contiguous records. We identify each firm-year’s CEO from ExecuComp’s annual CEO flag and drop the small number of firm-years with more than one executive so flagged. Appointment and departure years come from ExecuComp’s service dates; an executive who becomes

CEO in the second half of the year is assigned the following year as the first year of the spell. We classify CEOs as insiders or outsiders using prior tenure with the firm at the time of appointment: insiders are CEOs who are founders or who have more than one year of prior experience at the firm, and all other CEOs are outsiders. This threshold follows Parrino (1997).

**Forced and voluntary turnover.** A succession is classified as forced when the departing CEO's exit is coded as a dismissal in Gentry et al. (2021), and as voluntary when the CEO is replaced without a dismissal code. The database codes each departure from contemporaneous press reports, classifying it as a dismissal when the board removed the executive rather than the executive leaving of their own accord. Because press coverage understates board-initiated exits that are presented as retirements, this classification is conservative. The database's coverage ends in 2019, which sets the end of our sample window.

**Final sample.** Following Pérez-González (2006), we drop CEO spells lasting a single year at the time of turnover, as these are likely interim appointments. The final sample covers fiscal years 1995 through 2019 and contains 38,850 firm-year observations on 3,158 distinct firms and 6,508 CEO spells. For estimation we export one observation per firm-year containing operating ROA, the market-adjusted return, their industry-year benchmark means and industry-adjusted values, the insider indicator, the forced-turnover indicator, an indicator for a new CEO's first year, and CEO tenure. All regressions and moments used in estimation are computed from this file, on the simulated panel and the data identically.

## B Learning Dynamics: Proof of Proposition 1

**The filtering problem.** Stack the latent states  $q_t = (\theta_{it} - \mu_h, g_{nt})'$  and the signals  $z_t = (u_{jt} - (1 - \rho)\mu_h, s_{nt}^g)'$ . From (4), (3), (6), and (7), the system has the linear state-space form

$$q_{t+1} = F q_t + w_{t+1}, \quad z_t = H q_t + e_t, \quad (21)$$

$$F = \begin{pmatrix} \phi & 0 \\ 0 & \phi_g \end{pmatrix}, \quad H = \begin{pmatrix} 1 - \rho & 1 \\ 0 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} \sigma_\eta^2 & 0 \\ 0 & \sigma_\zeta^2 \end{pmatrix}, \quad R_e = \begin{pmatrix} \sigma_\varepsilon^2 & 0 \\ 0 & \sigma_\omega^2 \end{pmatrix},$$

with  $w_t \sim N(0, Q)$  and  $e_t \sim N(0, R_e)$  mutually independent. Let  $\hat{q}_{t|t-1}$  and  $\Sigma_{t|t-1}$  denote the one-step-ahead mean and covariance of  $q_t$  given the signal history. The Kalman recursions

are

$$r_t = z_t - H\hat{q}_{t|t-1}, \quad S_t = H\Sigma_{t|t-1}H' + R_e, \quad (22)$$

$$\hat{q}_{t|t} = \hat{q}_{t|t-1} + \Sigma_{t|t-1}H'S_t^{-1}r_t, \quad \Sigma_{t|t} = \Sigma_{t|t-1} - \Sigma_{t|t-1}H'S_t^{-1}H\Sigma_{t|t-1}, \quad (23)$$

$$\hat{q}_{t+1|t} = F\hat{q}_{t|t}, \quad \Sigma_{t+1|t} = F\Sigma_{t|t}F' + Q. \quad (24)$$

Because  $\phi, \phi_g < 1$ , the state is stationary and the Riccati iteration implied by (23)–(24) converges to a unique fixed point  $\Sigma^*$  from any initial covariance (Anderson and Moore, 1979). Steady-state learning evaluates the recursions at  $\Sigma^*$ , with stationary gain

$$K = \Sigma^*H'(H\Sigma^*H' + R_e)^{-1} \equiv \begin{pmatrix} K_u & K_g \\ L_u & L_g \end{pmatrix}. \quad (25)$$

whose first row  $(K_u, K_g)$  collects the loadings of the board's belief about  $\theta$  on the performance and benchmark signals, and whose second row  $(L_u, L_g)$  collects those of its belief about  $g$ . Write  $p_\theta = \Sigma_{11}^*$ ,  $p_g = \Sigma_{22}^*$ ,  $p_{\theta g} = \Sigma_{12}^*$ , and define the signal loadings

$$a_1 \equiv (1 - \rho)p_\theta + p_{\theta g}, \quad a_2 \equiv (1 - \rho)p_{\theta g} + p_g, \quad (26)$$

which are the covariances of the performance innovation  $u$  with the two latent states given the board's information. The  $\theta$ -row of (25) is then

$$\begin{pmatrix} K_u & K_g \end{pmatrix} = \begin{pmatrix} a_1 & p_{\theta g} \end{pmatrix} \begin{pmatrix} (1 - \rho)a_1 + a_2 + \sigma_\varepsilon^2 & a_2 \\ a_2 & p_g + \sigma_\omega^2 \end{pmatrix}^{-1}. \quad (27)$$

**The updating rule.** The  $\theta$ -component of the prediction step in (24), evaluated at the stationary gain and with the type mean added back, reads  $\tilde{\theta}_{it+1} - \mu_h = \phi[(\tilde{\theta}_{it} - \mu_h) + K_u r_{u,t} + K_g r_{g,t}]$ , where  $r_{u,t} = u_{jt} - \hat{u}_{jt}$  and  $r_{g,t} = s_{nt}^g - \hat{g}_{nt}$  are the two components of the innovation  $r_t$  in (22). Defining the benchmark-adjustment share  $\psi_g \equiv -K_g/K_u$  and factoring  $K_u$  out of the bracket yields equation (8).

**Signs of the gains.** The sign claims in part (i) follow from the signs of the stationary covariance, which we establish first.

**Lemma 1.** *Suppose  $\phi_g \leq \phi$ . At the stationary solution,  $p_{\theta g} < 0$ ,  $a_1 > 0$ , and  $a_2 > 0$ .*

PROOF: Because the two signals are conditionally independent, the update (23) can be computed sequentially: first on the benchmark signal  $s^g$  (loading  $(0, 1)$ ), then on the performance innovation  $u$  (loading  $(1 - \rho, 1)$ ). We show that the properties  $\{p_{\theta g} \leq 0, a_1 > 0, a_2 > 0\}$

are preserved by each step of the filter iteration; since the iteration converges to the unique fixed point from the diagonal initialization  $\Sigma_0 = \text{diag}(\sigma_\theta^2, \sigma_\zeta^2/(1 - \phi_g^2))$ , which satisfies them, the fixed point inherits them.

*Benchmark update.* With loading  $(0, 1)$  and noise  $\sigma_\omega^2$ , the update maps  $p_g \mapsto p_g \sigma_\omega^2 / (p_g + \sigma_\omega^2)$ ,  $p_{\theta g} \mapsto p_{\theta g} \sigma_\omega^2 / (p_g + \sigma_\omega^2)$ , and  $p_\theta \mapsto p_\theta - p_{\theta g}^2 / (p_g + \sigma_\omega^2)$ . Direct computation gives  $a'_2 = a_2 \sigma_\omega^2 / (p_g + \sigma_\omega^2) > 0$  and  $a'_1 = a_1 + (-p_{\theta g}) a_2 / (p_g + \sigma_\omega^2) \geq a_1 > 0$ ; the sign of  $p_{\theta g}$  is preserved.

*Performance update.* With loading  $(1 - \rho, 1)$  and noise  $\sigma_\varepsilon^2$ , write  $D = (1 - \rho)a_1 + a_2 + \sigma_\varepsilon^2 > 0$ . The update maps  $p_{\theta g} \mapsto p_{\theta g} - a_1 a_2 / D < 0$  (strictly, since  $a_1, a_2 > 0$ ), and direct computation gives  $a_1 \mapsto a_1 \sigma_\varepsilon^2 / D > 0$  and  $a_2 \mapsto a_2 \sigma_\varepsilon^2 / D > 0$ .

*Prediction.* The prediction step maps  $p_{\theta g} \mapsto \phi \phi_g p_{\theta g} \leq 0$  and, using  $\phi_g \leq \phi$ ,  $a_1 \mapsto (1 - \rho)(\phi^2 p_\theta + \sigma_\eta^2) + \phi \phi_g p_{\theta g} \geq \phi \phi_g a_1 + (1 - \rho)\sigma_\eta^2 > 0$ . For  $a_2$ , the prediction step gives  $a_2 \mapsto \phi_g^2 a_2 + (1 - \rho)\phi_g(\phi - \phi_g) p_{\theta g} + \sigma_\zeta^2$ , which is positive whenever  $\sigma_\zeta^2 \geq (1 - \rho)\phi_g(\phi - \phi_g) |p_{\theta g}|$ ; this inequality holds automatically when  $\phi_g = \phi$ , and we verify that it holds at the stationary solution for every parameter configuration considered in the paper. ■

Given Lemma 1, part (i) follows from (27): writing  $\det S > 0$  for the determinant of the innovation variance,

$$K_u = \frac{a_1(p_g + \sigma_\omega^2) - p_{\theta g} a_2}{\det S} > 0, \quad K_g = \frac{p_{\theta g}[(1 - \rho)a_1 + a_2 + \sigma_\varepsilon^2] - a_1 a_2}{\det S} < 0, \quad (28)$$

since every term in the numerator of  $K_u$  is positive and every term in the numerator of  $K_g$  is negative. Hence  $K_u > 0$  and  $\psi_g = -K_g/K_u > 0$ , establishing part (i).

**Limits.** As  $\sigma_\omega \rightarrow \infty$ , the benchmark signal carries no information:  $K_g \rightarrow 0$  at rate  $\sigma_\omega^{-2}$  while  $K_u$  converges to the positive gain of the one-signal filter, so  $\psi_g \rightarrow 0$ . As  $\sigma_\omega \rightarrow 0$ , the benchmark reveals  $g_{nt}$  exactly. The board's information is then equivalent to  $(u_{jt} - s_{nt}^g, s_{nt}^g)$ , where  $u_{jt} - s_{nt}^g = (1 - \rho)\theta_{it} + \varepsilon_{jt}$  is a signal about the CEO alone. Updating on it with coefficient  $k$  and using  $\hat{u}_{jt} = (1 - \rho)\tilde{\theta}_{it} + \hat{g}_{nt}$ ,

$$k[(u_{jt} - s_{nt}^g) - (1 - \rho)\tilde{\theta}_{it}] = k[(u_{jt} - \hat{u}_{jt}) - (s_{nt}^g - \hat{g}_{nt})], \quad (29)$$

so the coefficients on the performance and benchmark surprises are equal and opposite:  $K_g = -K_u$  and  $\psi_g = 1$ . ■

## C Moment Construction and Estimation Details

This appendix defines each targeted moment, the auxiliary regressions and samples from which they are computed, the weight matrix, the simulation and numerical routine, and the standard-error formulas. Every moment is computed by the same code on the data and on each simulated panel, so any approximation in a moment’s construction is shared between the two and nets out of the comparison.

**Inputs and timing.** Performance enters the moments in industry-adjusted form: profitability is operating ROA net of the industry-year universe mean, and returns are market-adjusted returns net of their industry-year mean, as described in Appendix A. In the hiring and firing regressions (19)–(20), the information set is dated relative to the decision year  $t$ : recent performance  $R^{\text{rec}}$  is the firm-specific market-adjusted return in year  $t - 1$ , older performance  $R^{\text{old}}$  is the average of years  $t - 2$  and  $t - 3$ , and the industry return  $R^{\text{ind}}$  is the industry-year mean of market-adjusted returns in year  $t - 1$ . All three covariates are standardized to unit variance within the estimation sample, so the targeted coefficients are responses to a one-standard-deviation movement.

**Profitability dynamics and variances (9 moments).** The pooled autocorrelations are the two slope coefficients from a regression of operating ROA on its first and second lags with a constant only; we deliberately omit fixed effects here to avoid the short-panel bias they would induce, and instead let the permanent firm component enter the moment on both the data and the simulation side. Mean ROA is the sample mean. The variance of the firm effect is the variance across firms of firm-average ROA. The within-spell variance is the variance of industry-adjusted profitability around CEO-spell means. The one-year and two-year change variances are the variances of  $y_{jt}^{fs} - y_{jt-1}^{fs}$  and  $y_{jt}^{fs} - y_{jt-2}^{fs}$  within CEO spells; differencing removes the permanent firm component, and the pair of horizons separates transitory noise from persistent movements in quality. The between-CEO variances by type are computed from spell-level averages of industry-adjusted profitability net of the firm’s average: the variance of these spell means across CEOs, separately for insider and outsider spells.

**Insider premium profile (12 moments).** The premium regression (18) is estimated on all CEO-years with firm and year fixed effects. The targeted coefficients are the six insider-by-tenure-bin interactions (bins 1, 2, 3, 4–6, 7–9, 10+), the coefficient on the turnover indicator (the succession dip), and the five baseline tenure-bin main effects (bins 2 through 10+; bin 1 is the base), which pin the tenure profile of performance under outsider CEOs.

**Hiring and firing margins (8 moments).** The hiring regression (20) is estimated on succession years only: the dependent variable is an indicator that the incoming CEO is an insider, the covariates are the departing regime’s performance measures defined above, and the regression includes firm and year fixed effects. The targeted moments are the three performance coefficients and the unconditional insider share of successions. The firing regression (19) is estimated on all CEO-years with firm and year fixed effects; the dependent variable is the forced-turnover indicator. The targeted moments are the three performance coefficients and the unconditional forced-turnover rate per firm-year.

**Industry factor (3 moments).** Let  $\bar{y}_{Jt}$  denote the industry-year universe mean of operating ROA. After removing industry means, the lag-one and lag-two autoregression coefficients of  $\bar{y}_{Jt}$  and its variance are targeted. In the data,  $\bar{y}_{Jt}$  blends the industry factor with the averaged firm-level components; the same blend arises in the simulated panel, where the moment is computed from the simulated industry means in exactly the same way, so the contamination is common to both sides and the moments remain informative about  $(\phi_g, \sigma_\zeta)$  jointly with  $\rho$ .

**Weighting, simulation, and the objective.** The weight matrix is diagonal, with each moment weighted by the inverse of its bootstrapped variance; we use the diagonal rather than the full inverse covariance matrix to avoid the small-sample bias of fully optimal weighting (Altonji and Segal, 1996). The bootstrap resamples firms with replacement, so the variances are clustered at the firm level; for the three industry-factor moments, whose effective variation is at the industry-year level, the bootstrap understates sampling error, and we replace their variances with industry-clustered bootstrap counterparts. The simulated moments are averaged over  $S = 20$  simulated economies. The criterion is minimized by a multistart array: each task runs a particle-swarm global stage, a pattern-search refinement, and a derivative-based polish from a distinct starting point, and the reported estimate is the best across tasks.

**Standard errors.** Standard errors account for simulation error and for the covariance across moments. Writing  $G$  for the Jacobian of the simulated moments in the parameters (computed by central finite differences under common random numbers),  $W$  for the diagonal weight matrix used in estimation, and  $\Sigma$  for the full bootstrapped covariance matrix of the empirical moments, the estimator’s asymptotic variance is the sandwich:

$$\widehat{\text{avar}}(\hat{\theta}) = \left(1 + \frac{1}{S}\right) (G'WG)^{-1} G'W \Sigma WG (G'WG)^{-1} \quad (30)$$

The factor  $(1 + 1/S)$  accounts for simulation noise.  $\Sigma$  is clustered at the firm level with the industry-cluster override described above.

## D First Stage: Voluntary Turnover

Table 10: First Stage: Voluntary Separation Hazard

| Coefficients                      |                       | Implied annual hazard $\delta(\tau)$ |       |
|-----------------------------------|-----------------------|--------------------------------------|-------|
| $\omega_0$ (constant)             | -3.902                | $\tau = 1$                           | 0.025 |
| $\omega_1$ (tenure)               | 0.259                 | $\tau = 5$                           | 0.053 |
| $\omega_2$ (tenure <sup>2</sup> ) | -0.0116               | $\tau = 10$                          | 0.089 |
| $\omega_3$ (tenure <sup>3</sup> ) | $1.43 \times 10^{-4}$ | $\tau = 15$                          | 0.105 |
|                                   |                       | $\tau = 20$                          | 0.099 |
|                                   |                       | $\tau = 30$                          | 0.064 |

Notes: Maximum-likelihood estimates of the logistic hazard (31) on the CEO-year panel. The right panel evaluates the estimated hazard at selected tenures.

Voluntary separation is modeled as an exogenous hazard in CEO tenure, estimated in a first stage and held fixed in the structural estimation. In each year an incumbent of tenure  $\tau$  departs voluntarily with probability

$$\delta(\tau) = \frac{1}{1 + \exp(-(\omega_0 + \omega_1\tau + \omega_2\tau^2 + \omega_3\tau^3))}, \quad (31)$$

and the same hazard applies in the model’s solution and in the simulated panels. Treating this margin as exogenous reflects its role in the model: the board’s decision problem concerns dismissal and appointment, and voluntary departures (retirements and outside opportunities) arrive independently of the board’s beliefs.

In the data, a voluntary exit is a CEO-year that is the last observed year of the spell, at a firm that remains in the panel afterward, and that carries no dismissal flag; requiring the firm to continue distinguishes a CEO departure from the firm itself exiting the sample, and the dismissal flag routes forced exits to the model’s endogenous firing margin. We estimate (31) by maximum likelihood on the CEO-year panel. Parameter estimates and the implied hazard profile are reported in Table 10. The hazard rises from about 2.5% in a CEO’s first year to a peak of roughly 10% around fifteen years of tenure and declines thereafter, consistent with retirement timing concentrating in the second decade of a spell and with very long-tenured CEOs (disproportionately founders) being unlikely to leave.

## E Sensitivity of Parameter Estimates to Moments

Table 11: Sensitivity of parameter estimates to targeted moments

| Parameter            | First                   |                   | Second                |                   | Third                   |                   |
|----------------------|-------------------------|-------------------|-----------------------|-------------------|-------------------------|-------------------|
|                      | Moment                  | $\tilde{\Lambda}$ | Moment                | $\tilde{\Lambda}$ | Moment                  | $\tilde{\Lambda}$ |
| $\bar{t}$            | Mean profitability      | −0.49             | Hiring on older ret.  | 0.37              | Premium, ten. 2         | 0.35              |
| $\sigma_\iota$       | Var. firm effect        | −0.92             | Profit. autocorr. (2) | −0.15             | Profit. autocorr. (1)   | −0.11             |
| $\rho$               | Profit. autocorr. (1)   | −0.41             | Var. 1-yr change      | 0.35              | Profit. autocorr. (2)   | 0.33              |
| $\sigma_\varepsilon$ | Var. 2-yr change        | −0.30             | Var. 1-yr change      | −0.30             | Within-spell var.       | −0.29             |
| $\phi$               | Premium, ten. 2         | 0.53              | Firing on recent ret. | −0.50             | Premium, ten. 4–6       | 0.26              |
| $\sigma_\eta$        | Premium, ten. 2         | −0.49             | Premium, ten. 3       | −0.30             | Hiring on older ret.    | −0.29             |
| $\psi_g$             | Firing on industry ret. | −0.71             | Var. industry mean    | −0.42             | Firing on older ret.    | 0.33              |
| $\kappa$             | Hiring on older ret.    | −0.56             | Hiring on recent ret. | −0.45             | Var. industry mean      | 0.41              |
| $\mu_I$              | Hiring on recent ret.   | 0.52              | Premium, ten. 2       | −0.50             | Premium, ten. 4–6       | 0.30              |
| $c$                  | Premium, ten. 2         | 0.48              | Hiring on older ret.  | 0.42              | Premium, ten. 4–6       | −0.40             |
| $\pi$                | Hiring on older ret.    | −0.53             | Hiring on recent ret. | 0.33              | Firing on recent ret.   | −0.32             |
| $\sigma_\nu$         | Var. industry mean      | −0.40             | Premium, ten. 4–6     | 0.34              | Premium, ten. 10+       | 0.34              |
| $\phi_g$             | Ind. autocorr. (1)      | −0.57             | Ind. autocorr. (2)    | −0.53             | Profit. autocorr. (2)   | −0.19             |
| $\sigma_\zeta$       | Var. industry mean      | −0.77             | Ind. autocorr. (2)    | 0.18              | Hiring on industry ret. | 0.18              |
| $\gamma$             | Premium, ten. 7–9       | −0.48             | Hiring on older ret.  | −0.35             | Firing on industry ret. | 0.33              |

Standardized sensitivities  $\tilde{\Lambda}_{pm} = \Lambda_{pm} s_m / s_p$  from (32), evaluated at the estimated parameter vector with the estimation weight matrix: the movement of parameter  $p$ , in units of its standard error, induced by a one-standard-error perturbation of moment  $m$ . For each parameter the table reports the three largest entries in absolute value. Moment labels: “Profit. autocorr.” are the pooled autocorrelations of industry-adjusted profitability; “Var. 1-yr (2-yr) change” the within-spell variances of one- and two-year profitability changes; “Var. firm effect” the across-firm variance of average profitability; “Ind. autocorr.” and “Var. industry mean” the autocorrelations and variance of the industry benchmark; “Premium, ten.  $b$ ” the insider premium coefficient for tenure bin  $b$ ; “Firing (Hiring) on recent/older/industry ret.” the coefficients of the firing (hiring) regressions on recent, longer-lagged, and industry returns.

This appendix quantifies the dependence of the parameter estimates on the targeted moments using the sensitivity measure of Andrews et al. (2017). Let  $G$  denote the Jacobian of the simulated moments with respect to the parameters and  $W$  the weight matrix, both evaluated at the estimated parameter vector. The sensitivity matrix is:

$$\Lambda = -(G'WG)^{-1}G'W \quad (32)$$

Entry  $\Lambda_{pm}$  measures the local effect of a perturbation of moment  $m$  on the estimate of parameter  $p$ : it answers the question of which features of the data are responsible for each estimate. To place entries on a common scale, we standardize by the precision of the moments and of the estimates, reporting  $\tilde{\Lambda}_{pm} = \Lambda_{pm} s_m / s_p$ , where  $s_m$  is the standard error of moment  $m$  and  $s_p$  the standard error of parameter  $p$ . A standardized entry of 0.5 thus indicates that

a one-standard-error perturbation of the moment moves the parameter estimate by half of its standard error.

Table 11 reports, for each parameter, the three most influential moments and their standardized sensitivities. The patterns line up closely with the identification discussion in Section 4. The dispersion of the permanent firm component is determined almost entirely by the across-firm variance of profitability, and its mean by average profitability. The idiosyncratic shock volatility  $\sigma_\epsilon$  draws essentially all of its information from the three within-spell variance moments, the variances of profitability and of its one- and two-year changes, while firm persistence  $\rho$  combines the pooled autocorrelations with the change variances, exactly the division of labor described in the text. The industry parameters are pinned by the industry block: the autocorrelations of the industry benchmark for  $\phi_g$  and its variance for  $\sigma_\zeta$ . On the CEO side, the persistence of productivity  $\phi$  and the innovation volatility  $\sigma_\eta$  load on the tenure profile of the insider premium, together with the responsiveness of firing to recent performance: both are manifestations of the speed at which productivity, and hence beliefs, evolve. The type mean  $\mu_I$  draws on the premium profile at intermediate and long tenures, where productivity has reverted toward its type mean, and continuity  $\kappa$  loads on the dependence of hiring choices on the predecessor’s track record, its distinguishing implication. The attribution parameter  $\psi_g$  is dominated by the response of firing to industry returns, the moment that directly measures the extent to which industry conditions enter personnel decisions.

Because the parameters are estimated jointly, the sensitivity matrix also reflects the covariation of estimates through  $(G'WG)^{-1}$ : a moment can influence a parameter it does not directly inform when the two are connected through a third parameter. The appearance of the industry variance among the sensitivities of several non-industry parameters is of this kind: it anchors the scale of industry variation on which the attribution and screening channels operate. Similarly, the preference parameters draw on the hiring and firing margins as a whole rather than on any single moment, reflecting that they shift the levels of both margins jointly.

## F Model Robustness

This appendix re-estimates the model under alternative maintained assumptions. Each variation relaxes one restriction imposed in Section 3, re-estimates the full parameter vector against the same moments with the same weighting matrix and the same simulation draws, and reports the result against the baseline of Table 5. The purpose is to establish which of our conclusions depend on modelling conveniences and which do not.

## F.1 Non-Stationary Learning

Section 3 holds the board’s posterior covariance at its stationary value at every tenure. Here we relax that restriction and re-estimate, letting the posterior variance depend on elapsed tenure and on the type of the executive appointed. Insiders then enter better informed than outsiders, with an entry variance of 0.66 of  $\sigma_\theta^2$  against 1.00, and the two paths converge as tenure accumulates. The filter introduces no additional parameters.

Table 12: Estimates under Stationary and Non-Stationary Learning

| Parameter                              | Description                         | Baseline | Non-stationary |
|----------------------------------------|-------------------------------------|----------|----------------|
| <i>CEO productivity and succession</i> |                                     |          |                |
| $\phi$                                 | Productivity persistence            | 0.980    | 0.981          |
| $\sigma_\eta$                          | Productivity-innovation s.d.        | 0.007    | 0.007          |
| $\mu_I$                                | Insider productivity premium        | 0.006    | 0.007          |
| $\kappa$                               | Continuity (as fraction of $\phi$ ) | 0.328    | 0.329          |
| $\lambda = \kappa\phi$                 | Implied inheritance                 | 0.321    | 0.323          |
| <i>Board information</i>               |                                     |          |                |
| $\psi_s$                               | Screening precision                 | 0.306    | 0.283          |
| $\psi_g$                               | Industry-performance filtering      | 0.625    | 0.657          |
| <i>Firm profitability</i>              |                                     |          |                |
| $\bar{\iota}$                          | Firm-effect mean                    | 0.104    | 0.102          |
| $\sigma_\iota$                         | Firm-effect s.d.                    | 0.098    | 0.099          |
| $\rho$                                 | Operational persistence             | 0.512    | 0.493          |
| $\sigma_\varepsilon$                   | Transitory shock s.d.               | 0.061    | 0.061          |
| $\phi_g$                               | Industry-factor persistence         | 0.445    | 0.437          |
| $\sigma_\zeta$                         | Industry-factor innovation s.d.     | 0.017    | 0.017          |
| $c$                                    | Monetary turnover cost              | 0.0003   | 0.0004         |
| <i>Board preferences</i>               |                                     |          |                |
| $\pi$                                  | Non-pecuniary firing cost           | 0.075    | 0.082          |
| $\gamma$                               | Insider hiring preference           | 0.004    | 0.004          |
| Objective                              |                                     | 45.55    | 44.01          |

Notes: Both columns are estimated on the same 32 moments with the same weighting matrix and the same simulation draws, and both estimate the same 15 parameters: the non-stationary filter introduces no additional parameters. The baseline column reproduces Table 5. The objective is the weighted moment criterion, so the two values are directly comparable.

Table 12 reports the estimates. No parameter changes materially. The screening precision falls from 0.31 to 0.28 and the dismissal wedge rises from 0.075 to 0.082; every other estimate moves by less than the width of a standard error in Table 5. The preference for internal

appointment remains statistically indistinguishable from zero, and the fit is slightly better, though the improvement is small.

One quantity becomes a profile rather than a constant. Industry filtering  $\psi_g$  rises from roughly 0.55 in an executive’s first year to 0.66 at long tenure, because a board still uncertain about a new CEO attributes more of a common industry movement to that executive; the value reported in Section 5 is the long-tenure one. Given how little else moves, we retain the stationary gain in the baseline specification.

## F.2 Partial Information about Outside Candidates

The model gives boards no candidate-specific information about an external candidate. We relax that with a single parameter  $\psi_O \in [0, 1]$ : boards observe a noisy pre-hire signal about the outside candidate, the analogue of the screening signal they receive about an internal one, and  $\psi_O$  is the share of the variance of the outsider’s productivity that this signal resolves, so the board’s pre-hire posterior mean is distributed  $N(\mu_O, \psi_O \sigma_\theta^2)$ . At  $\psi_O = 0$  the model is the one estimated in the text. Parameters are held at their estimated values, the board re-solves its policy, and the dismissal probability defining the uninformed monitoring rule is recalibrated within each regime, so every column is compared to its own baseline. Table 13 reports  $\psi_O \in \{0.05, 0.15, 0.34\}$ ; at the largest value boards resolve as large a share of an outside candidate’s productivity variance as they resolve about an internal one, 27.9% through the screening signal and a further 6.2% through inheritance. The value of board information rises with  $\psi_O$  at both margins, and the ordering of the two margins is preserved throughout. Each row removes the board’s information at one margin, so the better informed the board is to begin with, the more its removal costs. The figures in the text are therefore conservative with respect to this assumption.

Table 13: Counterfactuals under partial information about outside candidates

|                                      | $\psi_O$     |        |        |        |
|--------------------------------------|--------------|--------|--------|--------|
|                                      | 0 (baseline) | 0.05   | 0.15   | 0.34   |
| <i><math>\Delta</math> Value (%)</i> |              |        |        |        |
| Uninformed selection                 | −4.64        | −4.87  | −5.19  | −6.18  |
| Uninformed monitoring                | −4.10        | −4.48  | −4.82  | −5.40  |
| Uninformed at both margins           | −9.22        | −9.98  | −11.01 | −12.71 |
| <i><math>\Delta</math> ROA (%)</i>   |              |        |        |        |
| Uninformed selection                 | −5.54        | −5.97  | −6.36  | −7.38  |
| Uninformed monitoring                | −4.84        | −5.40  | −5.78  | −6.33  |
| Uninformed at both margins           | −10.98       | −11.94 | −13.10 | −14.95 |
| <i>Baseline regime</i>               |              |        |        |        |
| Insider share                        | 0.658        | 0.650  | 0.641  | 0.627  |
| Forced-turnover rate                 | 0.027        | 0.032  | 0.034  | 0.039  |

Notes: Each column is a separate regime in which the board resolves share  $\psi_O$  of the variance of an outside candidate's productivity before hiring. Parameters are held at the estimates of Table 5; the board re-solves its policy, and the dismissal probability defining the uninformed monitoring rule is recalibrated to reproduce that regime's own forced-turnover rate. Value and ROA changes are measured against the baseline of the same column, and the insider share and forced-turnover rate reported are those of each column's own baseline regime rather than of the rows above. The value and ROA changes in the first column reproduce Panel A of Table 6.